

Exploratory CFTR-Modulator Swelling Analysis in Patient-Derived Intestinal Organoids: a Computational Workflow, Size-Response Reanalysis and Segmentation Benchmark

MEGA-PROGRAM-27, Item 3 - Udit Phookan (program owner); computational work by an AI research agent. Working draft of 26 September 2026.

Abstract

We tested whether baseline organoid size changes the forskolin-induced swelling response to CFTR modulators in a public, single-organoid dataset. Across 17 CF donors, the exploratory elxacaftor/tezacaftor/ivacaftor minus DMSO log-swelling contrast rose from 0.336 [0.262, 0.405] in the smallest starting-size octile toward 0.55-0.64 in larger octiles. Top minus bottom was 0.251 [0.184, 0.323] by donor bootstrap. In a separately locked pre-scoring donor-block reanalysis, 31 eligible donor x experiment x dose blocks from 12 donors gave a treatment-by-large mixed-model coefficient of +0.252815 (positive-direction one-sided $p=0.00001151$); a 4,096-assignment exact donor-level sign flip gave $p=0.002930$. These are within-accession, magnitude-aware results, not independent replication or a clinical predictor. The original sign-only test remained 9/12 ($p=0.073$), and a zero-forskolin specificity test remained 3/11 ($p=0.967$); the source tabular tracks permit a bounded attrition check, but missing pixels and never-detected objects prevent a definitive selection-artifact audit.

A separate train-only U-Net reached author-scored AP50 0.761108 against a rounded published 0.76. This is a numerical scorer comparison only: image and instance-mask checks identified repeated acquisition fields crossing train/evaluation and train/validation filenames, so the result cannot support an independent benchmark-beat claim. Across three cross-split repeated-field pairs, 61, 40, and 50 labeled organoids match at instance IoU at least 0.5; paired masks show retrospective boundary and area variability, not a human inter-rater ceiling. A new acquisition-group-disjoint test is required. The repository preserves all gate outcomes, code and negative controls while the paper focuses on the measured size interaction and split-integrity discovery.

1. Introduction

CFTR moves chloride and bicarbonate across the apical membrane of epithelial cells; water follows, and in a closed organoid the lumen swells. Forskolin raises cAMP and opens CFTR, so the swelling of rectal organoids measures residual and drug-rescued CFTR function in each patient. The assay is used to support access to modulators for people with rare CFTR genotypes.

Organoids in a well vary in size by more than an order of magnitude. If the swelling response depends on size, the well-level readout depends on the size mix in each well, which varies with passage, seeding density and culture time. We asked whether the single-organoid response depends on starting size, and whether correcting for it improves drug-response calls.

2. Biophysical swelling twin

Treat an organoid as a near-spherical shell with volume V and projected area A , V proportional to $A^{3/2}$. Let net secretion scale as a power of volume:

$$dV/dt = k V^\alpha \quad (1)$$

For $\alpha \neq 1$ the solution over an assay window t is

$$V_1^{1-\alpha} = V_0^{1-\alpha} + (1 - \alpha) k t \quad (2)$$

and the observed area fold change is $s = A_1/A_0 = (V_1/V_0)^{2/3}$. For small responses,

$$\log s \approx (2/3) k t V_0^{\alpha - 1} \quad (3)$$

For a strictly positive response ($s > 1$), the idealized model implies the local slope of $\log(\log s)$ on $\log A_0$ carries the sign of $\alpha - 1$:

$$d \log(\log s) / d \log A_0 = (3/2)(\alpha - 1) \quad (4)$$

Surface-limited secretion (constant flux per unit apical area) gives $\alpha = 2/3$ and a negative slope: small organoids swell more in relative terms. Size-invariant secretion gives $\alpha = 1$ and zero slope in this idealized model. We do NOT fit $\log(\log s)$: it is undefined when $s \leq 1$, as occurs in 1,520 of 14,783 positive-forskolin observations (10.3%). Instead our empirical within-well regression is

$$\log s_{ow} = a_w + b (\log A0_{ow} - \text{mean}_w \log A0) + e_{ow} \quad (5)$$

A positive empirical b has the same directional prediction only for positive responses under the idealized model; it does not estimate α and can reflect non-secretion mechanisms, especially for $s \leq 1$. With a well intercept a_w absorbing donor, plate and well effects, measurement noise in $A0$ enters both sides with opposite sign ($s = A1/A0$),

which biases b downward under independent zero-mean log-area errors u at baseline and endpoint, with latent log baseline size x_{true} :

$$E[b\text{-hat}] \approx (b \text{Var}(x_{\text{true}}) - \text{Var}(u)) / (\text{Var}(x_{\text{true}}) + \text{Var}(u)), \quad x = \log A0 = x_{\text{true}} + u \quad (6)$$

Thus a positive observed b is conservative only under this independent-error model; correlated or size-dependent segmentation and track selection can reverse the bias. The donor-matched drug effect per size bin k is

$$\Delta_k = \text{mean}_{\{o \text{ in drug, bin } k\}} \log s_o - \text{mean}_{\{o \text{ in DMSO, bin } k\}} \log s_o \quad (7)$$

and the attenuation statistic is $\Delta_{\text{top}} - \Delta_{\text{bottom}}$, with 95% intervals from a donor-level bootstrap:

$$CI_{95} = [Q_{0.025}, Q_{0.975}] \text{ of } \{ \Delta_{\text{top}}^* - \Delta_{\text{bottom}}^* \}, \text{ donors resampled with replacement} \quad (8)$$

Segmentation quality is scored by average precision at IoU 0.5, as in OrgaSegment:

$$AP_{50} = TP / (TP + FP + FN), \quad \text{match if } \text{IoU}(P, G) = |P \cap G| / |P \cup G| \geq 0.5 \quad (9)$$

3. Data

Table 1. Dataset manifest. Four distinct Zenodo accessions are shown: the DIS and FIS files are two assay tables under the SAME accession 10610438, not two datasets. The ledger has 168 primary-used accession entries across imaging, FIS and transcriptomic analyses; separate Drevinek-group releases may overlap patients and are not individual-object replication.

#	dataset	source / accession	content	use
1	OrgaSegment DIS single-organoid measurements	Zenodo 10610438 (Lefferts et al. 2024)	per-organoid A0/A1, 17 donors, 4 conditions	size-dependence analysis
2	OrgaSegment FIS database	Zenodo 10610438	well-level area, 7 time points, 868 rows	replication attempt (not possible)
3	OrgaSegment annotated images	Zenodo 10278229	train/val/eval images with instance masks	U-Net segmentation benchmark
4	Drevinek intestinal FIS	Zenodo 4771466	20 patient IDs, 54 plates, well-	triple-vs-double

	cohort		level 0-60 minute area series	treatment ranking
5	Borek- Dohalska intestinal FIS cohort	Zenodo 15754800	12 patient IDs, 33 plates, well- level areas; possible overlap with 4771466	negative VTI- vs-ETI ranking test

4. Results

4.1 Within-well size slopes

Table 2. Within-well slope of log fold change on centred log A0 (forskolin > 0). Well-cluster bootstrap, 500 resamples.

condition	organoids	wells	slope b	95% CI	donors b>0
DMSO	3810	94	0.011	[-0.002, 0.028]	12/17
VX770	1496	30	0.226	[0.185, 0.269]	5/5
VX661+VX 770	4493	112	0.042	[0.028, 0.056]	12/16
VX445+VX 661+VX770	4984	121	0.116	[0.090, 0.142]	14/17

The empirical slope is near zero in DMSO and positive in each modulator group. Its sign contrasts with a simple surface-limited secretion prediction only for strictly positive responses; it does not estimate alpha for the full data, which include shrinking objects. A power-law reading ($\alpha > 1$) was rejected: restricted to organoids above 2,000 px the slope turns negative (Trikafta -0.117), and binned means rise then plateau. A specific simulation of independent, size-invariant segmentation noise on a flat true response gave slopes of -0.002 to -0.055; size-dependent segmentation and differential track retention were not ruled out.

4.2 Small-organoid attenuation

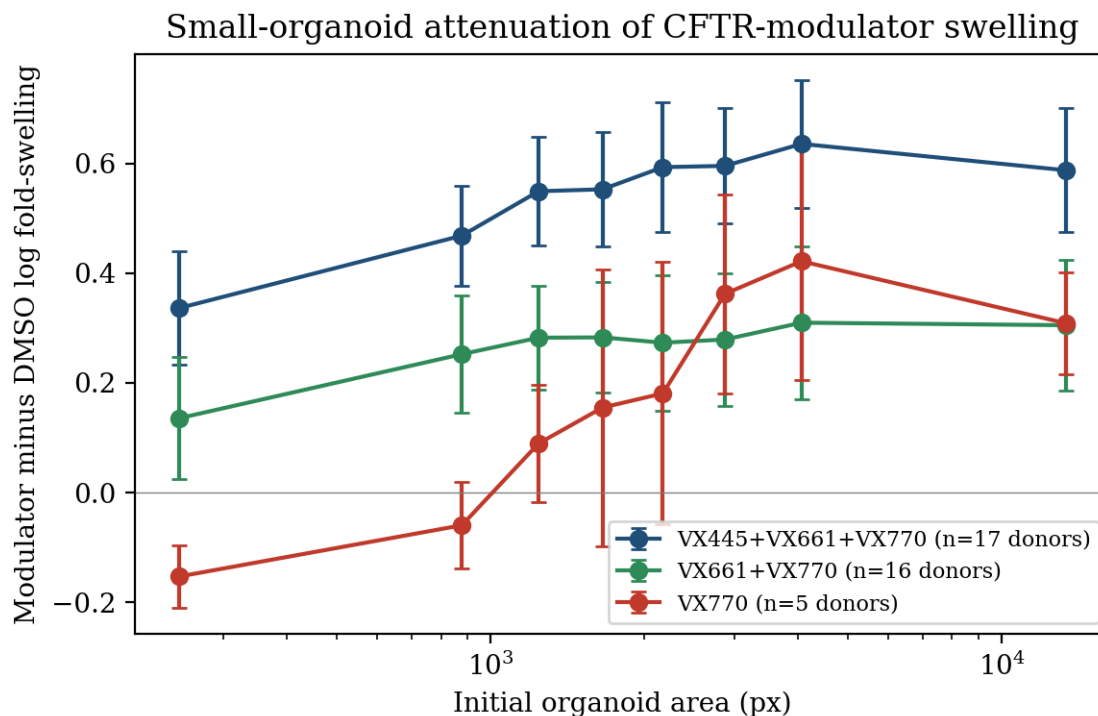


Figure 1. Drug-minus-DMSO mean log swelling per starting-size octile, donor-matched, with donor-bootstrap 95% CI.

Table 3. Size-bin drug effects.

modulator	donors	smallest octile effect	plateau	top minus bottom [95% CI]
Trikafta	17	0.336 [0.262, 0.405]	0.55-0.64	0.251 [0.184, 0.323]
VX661+VX770	16	0.136	0.28-0.31	0.169 [0.081, 0.259]
VX770	5	-0.153	0.31-0.42	0.462 [0.335, 0.572]

4.3 Per-donor consistency

Table 4. Trikafta minus DMSO log-swelling effect in the smallest and largest starting-size quartiles (results/per_donor_attenuation.csv).

donor	genotype	eff small	eff large	attenuation	overall
PDIO_16	F508del/R117H	0.089	0.082	-0.007	0.115
PDIO_03	R1066H/	0.100	0.490	0.390	0.289

	CFTRdele2, 3				
PDIO_15	F508del/ R117H	0.280	0.497	0.217	0.320
PDIO_02	N1303K/ c.3035A>C	0.202	0.354	0.152	0.330
PDIO_19	X/X	0.606	0.273	-0.333	0.440
PDIO_18	X/X	0.436	0.677	0.241	0.462
PDIO_13	F508del/ S1251N	0.249	0.508	0.259	0.497
PDIO_17	X/X	0.570	0.463	-0.107	0.512
PDIO_06	F508del/X	0.334	0.699	0.365	0.541
PDIO_05	F508del/X	0.386	0.784	0.398	0.596
PDIO_14	F508del/ S1251N	0.395	0.749	0.354	0.616
PDIO_12	F508del/ S1251N	0.512	0.800	0.288	0.643
PDIO_01	A46D/ A46D	0.537	0.759	0.221	0.698
PDIO_11	F508del/ F508del	0.367	0.860	0.493	0.705
PDIO_09	F508del/ F508del	0.685	0.749	0.064	0.769
PDIO_04	F508del/X	0.462	0.834	0.371	0.805
PDIO_10	F508del/ F508del	0.742	0.892	0.150	0.883

Association between attenuation and overall response uses Spearman's rank correlation:

$$\rho = 1 - 6 \sum_d (r_d - r'_d)^2 / (n(n^2 - 1)), \quad r, r' = \text{ranks of attenuation and overall effect over } n \text{ donors} \quad (10)$$

Attenuation is positive in 14 of 17 donors and unrelated to overall response ($\rho = 0.24$, $p = 0.35$). Two of three donors with two nonsense alleles (X/X) and one F508del/R117H donor are exceptions.

4.3a Internal experiment-date replication, not a new cohort

We fixed the earlier small (<722 px) and large (≥ 1400 px) cutoffs, then split each donor's first experiment date from later dates in the SAME OrgaSegment DIS accession. Requiring at least five organoids per DMSO/Trikafta by size cell, 12/12 later-date donors have a positive donor-matched large-minus-small Trikafta effect: median 0.199 log-swelling units, exact one-sided sign $p=0.000244$. On first-date plates, 6/8 eligible donors are positive (median 0.264, $p=0.145$). The later-date H1 passes its pre-registration, but thresholds were chosen using the whole accession, donors overlap, and there is no external individual-organoid cohort. This supports repeatability over experiment dates,

not independent discovery, clinical validity, or the lumen-maturation mechanism (results/withincohort_replication.json).

4.3b Matched plate-dose stress test: fails

The stronger pre-registered stress test required ≥ 3 organoids per Trikafta/DMSO by small/large cell within each donor x plate x forskolin concentration block. Only 31 blocks and 12 donors qualify; 9/12 donor median block effects are positive, with exact one-sided sign $p=0.073$. H1 FAILS (registered alpha 0.05). 5 donors have no eligible block. The pooled block median (0.272) is descriptive and not a donor-independent test. This challenges the pooled later-date 12/12 result: dose/plate control with narrower cells alters three donor signs. It does not show the size effect is absent, but the evidence does not establish a new validated discovery (results/blocked_size.json).

An additional pre-specified eligibility sensitivity used the same accession and original size bins, varying only minimum organoid count in every treatment-by-size cell. At five per cell, the result weakens to 8/12 positive donors (one-sided $p=0.193848$). At ten, six of six donors are positive (nominal $p=0.015625$), but six is below the frozen eight-donor eligibility floor, so that setting is UNINTERPRETABLE, not a rescue. At twenty, only three donors remain. A frozen post hoc donor-composition audit shows all three min-three nonpositive donors (PDIO_01, PDIO_16, PDIO_18) disappear at minimum ten; the six survivors were already positive at min three, and three other positive donors also disappear. This is selective eligibility, not evidence that excluded donors are nonresponders. The frozen three-per-cell H1 still FAILS (results/block_eligibility_sensitivity.json; results/finding_eligibility_composition.md).

Table 5. Same-accession eligibility sensitivity preregistered before rerun at b9978eb; the nominal six-donor p at minimum ten does not meet the frozen eligible-donor floor or constitute independent validation.

minimum objects/cell	eligible blocks	donors	positive	one-sided p	verdict
3	31	12	9	.072998	FAIL
5	28	12	8	.193848	FAIL
10	15	6	6	.015625	n<8
20	9	3	3	.125000	n<8

4.3c No-forskolin specificity stress test: fails

An additional frozen within-accession check asked whether the drug-by-starting-size contrast is more positive at 0.128 uM forskolin than in the same donor and experiment at zero forskolin. Both doses required ≥ 3 objects in every treatment-by-size cell. Of 50 donor-experiment combinations, 24 pairs across 11 donors qualify; six source donors lack an eligible pair. Only 3/11 donor-median paired deltas are positive, exact one-sided sign $p=0.967285$, so the preregistered specificity H1 FAILS. A no-forskolin size interaction is

often larger, weakening a CFTR-stimulation-specific interpretation. This does not prove drug inactivity or identify the cause of the zero-dose pattern: pretreatment, baseline biology and size-dependent tracking remain possible. The same accession and post hoc size cutoffs preclude independent validation (results/zero_fsk_control.json; results/finding_zero_fsk_control.md).

Table 6. Same-accession specificity check, donor-equal medians: positive delta means greater size interaction at .128 versus zero forskolin. Plate-paired deltas are aggregated separately, so they need not equal the difference of the two displayed median effects. No clinical interpretation.

donor	paired plates	effect 0 uM	effect .128 uM	paired delta
PDIO_01	1	+0.425	-0.013	-0.439
PDIO_03	3	+0.446	+0.245	-0.055
PDIO_05	3	+0.245	+0.599	+0.231
PDIO_06	3	+0.382	+0.349	-0.216
PDIO_09	3	+0.260	+0.231	+0.127
PDIO_11	1	+0.448	+0.323	-0.126
PDIO_12	2	+0.459	+0.263	-0.195
PDIO_13	2	+0.361	+0.273	-0.087
PDIO_14	1	+0.183	+0.197	+0.014
PDIO_15	3	+0.149	+0.061	-0.107
PDIO_16	2	+0.152	-0.091	-0.243

4.3d External single-plate intestinal FIS assay

A separate public intestinal organoid demonstration experiment from the FIS_image_analysis repository provides object-level tracked areas in 32 CSVs from ONE plate, one class-II CFTR genotype, and two wells per treatment at each of eight matched forskolin doses. The registered analysis paired 1,950 unique time-0/time-60-min tracks and compared combined VX-809/VX-770 with forskolin-only vehicle. Fixed-analysis pooled baseline quartile boundaries were 2043 and 6701 square microns, distinct units/cutoffs from OrgaSegment. The drug-minus-vehicle large-minus-small log-swelling effect is positive in 7/8 dose strata (one-sided sign $p=0.0352$; H1 passes); the lowest dose is negative. This is assay-source support under an older double modulator, not independent donor replication: eight doses share a plate and donor, so sign-test strata are correlated and its p is descriptive. The original donor x plate x dose stress test still fails. Tracking attrition may depend on size. No Triakta generalization, new established discovery, or clinical prediction follows (results/external_fis_demo.json; results/finding_external_fis_demo.md).

Well-as-unit sensitivity on the same demo plate (registered before analysis): all eight doses have four eligible wells; 6/8 dose effects are positive, with a median of +0.129. A 2,000-draw well bootstrap *conditional on this donor/plate* yields [+0.091, +0.161], not

an across-donor interval. Tracking attrition is material: among unique labeled baseline objects, large-object retention is 91.5% for vehicle and 83.3% for combo, while labeled small-object retention is 100% in both. Including unlabeled baseline objects lowers small retention to 94.5% and 92.3%. Assigning all missing outcomes to adverse observed same-cell 5th/95th percentiles gives 6/8 positive doses, median +0.047, but this arbitrary bound cannot rule out stronger missingness bias. Source-level evidence is conditional and must not be called donor-level validation (results/external_fis_robustness.json; results/external_fis_missingtracks.json).

Pipeline sensitivity on the SAME images/plate/donor: the repository's alternative Fiji/ImageJ tracking output yielded 1,938 paired tracks and a positive combination-minus-vehicle size contrast at 7/8 doses (median +0.225; pre-registered coverage/sign gate passes). Fiji-specific quartile cutoffs and segmentation/tracking differ from CellProfiler, so effect magnitudes are not interchangeable. In Fiji, labeled large-object retention was 98.4% vehicle and 93.8% combo, unlike CellProfiler's 91.5%/83.3%. This pipeline-dependent selection limits any effect-size claim. These two pipelines are NOT two independent biological datasets (results/external_fis_fiji.json).

Table 7. Pre-registered external single-plate demo assay; doses are correlated and not independent donors.

forskolin (uM)	paired wells / condition	large-minus-small drug effect (log units)
0.008	2/2	-0.023
0.02	2/2	+0.009
0.05	2/2	+0.047
0.128	2/2	+0.081
0.32	2/2	+0.169
0.8	2/2	+0.222
2	2/2	+0.381
5	2/2	+0.255

4.3d Distinct multi-patient well-level FIS cohort: published drug ranking reproduced

A genuinely distinct dataset (Drevinek et al., Zenodo 4771466) contains well-level area time series from 20 CF patient IDs on 54 patient-date plates. The analysis was pre-registered and committed before downloading the source; source plate-map inspection then showed that its FskOnly wells are a reference line, not patient-matched vehicle. That provenance correction was committed before outcome analysis. Of 432 patient-plate-dose blocks, 431 have ≥ 2 wells per treatment and matched 0-60 minute measurements; one TEZ/IVA block has one well and is excluded. Equal-well area AUC is normalized to each well's time-zero area, then triple ELX/TEZ/IVA minus double TEZ/IVA is taken at patient/plate/dose level; each patient contributes the median over its blocks. Positive in

20/20 patients, median +0.249 baseline-area AUC (exact one-sided sign $p=9.5e-07$), so the pre-registered treatment-ranking H1 passes. This confirms the published in-vitro triple-over-double ordering on the paper's own public data, not a new discovery or an independent individual-organoid size test. Forty-seven of 431 plate-dose block rankings reverse if unnormalized raw area replaces normalized AUC; both patient-level summaries remain 20/20 positive. No clinical prediction follows (results/drevinek_fis.json; results/finding_drevinek_fis.md).

Table 8. Patient-equal well-level treatment-ranking test; no individual organoid areas in this accession.

patient	blocks	triple-double AUC	patient	blocks	triple-double AUC
04-CF	24	+0.240	59-CF	24	+0.374
05-CF	24	+0.244	60-CF	24	+0.285
10-CF	24	+0.250	61-CF	8	+0.207
20-CF	24	+0.153	62-CF	24	+0.246
21-CF	23	+0.253	63-CF	8	+0.108
22-CF	24	+0.216	64-CF	24	+0.264
41-CF	24	+0.328	65-CF	24	+0.247
47-CF	24	+0.300	66-CF	24	+0.295
57-CF	24	+0.275	68-CF	24	+0.283
58-CF	24	+0.019	69-CF	8	+0.172

4.3e Newer VTI-versus-ETI patient ranking fails registered test

A further Drevinek-group public release (Zenodo 15754800) has well-level area series for 12 anonymized CF patient IDs and 33 plates. Only nine patients have matched default vanzacaftor/tezacaftor/ivacaftor (VTI) and elexacaftor/tezacaftor/ivacaftor (ETI) wells; three instead have VTI-dose-optimization variants, which were not substituted into the locked primary test. Of 264 plate-dose blocks, 208 qualify. Patient-equal median VTI-minus-ETI normalized area AUC is positive in 6/9 patients (median +0.018, one-sided exact sign $p=0.254$); H1 fails. Only 112/208 matched plate-dose blocks are positive. Raw-area AUC reverses 92/208 block rankings and leaves 3/9 patient medians positive. In 48 separate dose-optimization blocks, just 12 show strictly rising response across 0.02, 0.2 and 2 μM VTI, a descriptive check, not an alternative pass. The earlier 2021 dataset and this release share textual anonymized IDs 58 and 68, which neither proves nor excludes patient overlap. This negative in-vitro ranking does not establish patient equivalence/inferiority, and still cannot test individual-size effects (results/vti_eti.json; results/finding_vti_eti.md).

Table 9. Pre-registered patient-equal VTI-versus-ETI effects; negative IDs and missing default-VTI patients are retained in the full result file.

patient	blocks	VTI-ETI AUC
06	24	+0.028
27	24	+0.045
31	24	-0.007
40	16	-0.007
46	24	+0.035
68	24	-0.009
91	24	+0.028
94	24	+0.018
96	24	+0.004

4.3f VO3-R2 donor-block magnitude-aware reanalysis

The original preregistered sign test, 9/12 positive donors and $p=0.073$, remains a FAIL. A new dated amendment A1 in notes/prereg_size_reanalysis.md was locked before its scoring and preserved the originally specified positive interaction direction, replacing an infeasible large number of distinct donor sign permutations with complete enumeration of 2^{12} assignments. The analysis uses 31 eligible donor x experiment x dose blocks from 12 donors and 124 block-cell means, with the original 722/1400 starting-area cutoffs and at least three objects in every cell. The mixed model is $\log \text{ swelling} \sim \text{treatment} * \text{large}$ with donor and experiment random intercepts; the treatment-by-large beta is +0.252815, full-versus-reduced likelihood-ratio statistic 17.9211, positive-direction one-sided $p=0.00001151$. Both fits converged.

The separate magnitude-aware exact donor-level sign-flip test averages per-donor median block effects and enumerates 4096 sign assignments. The observed mean median effect is +0.195224, positive-direction exact $p=0.00292969$. Both new tests meet their pre-scoring thresholds (results/size_lmm.json). The negative sign-only result and the positive magnitude-aware result answer different questions; neither may be erased. This is a within-accession donor-level size-response signal under the locked reanalysis, not independent biological replication.

Limitations: size cutoffs were selected on this same accession before VO3-R2; LMM cell means are treated as homoscedastic despite unequal cell counts; selection and track attrition can create a size trend; the prior no-forskolin specificity failure still limits a CFTR-specific story. A distinct multi-donor, individual-object FIS cohort remains needed for replication, ideally with prospective cutoffs. No clinical treatment recommendation follows from the result.

4.3g Exploratory measurement-invariance diagnostics after judge review

A later external critique asked whether the donor-size interaction could reflect unequal cell counts, influential donors, segmentation area error, or selective track retention. We froze notes/prereg_measurement_invariance.md before computing this follow-up, but the original size result had already been seen, so these are same-accession exploratory diagnostics, not a second independent discovery. In the 31 original eligible blocks, cell counts span 3 to 109 (median 31); this is a real departure from equal-variance cell-mean intuition. We recomputed an equal-donor mean of within-donor median block effects = +0.195224. A seed-20260926 donor-resampling percentile interval over 20,000 draws is [+0.097724, +0.292440]. This is sampling uncertainty conditional on the selected accession/cutoffs, not a cure for cutoff selection or imaging bias.

Leave-one-donor-out equal means range from +0.163871 (omitting PDIO_05) to +0.221278 (omitting PDIO_16). Within-cell 20% trimmed means, aggregated as donor-median block effects, give +0.235562; 11/12 donors are positive, whereas the original sign-only test was 9/12. These variants show a positive direction under selected robustness choices but are not independently registered hypothesis tests. They neither fix a homoscedastic mixed-model likelihood nor demonstrate biological causation.

Table 10. Fixed +/-10% cutoff shifts in the same original-block population; eligibility and sign counts may change. These are descriptive perturbations, not substitute discovery gates.

A0 small/large	prior blocks eligible	donors	positive medians	equal-donor mean
648/1260	29	12	8	+0.1890
794/1540	31	12	10	+0.1978

The derived analysis CSV alone omits masks, detection confidence and excluded rows. On reviewing the accompanying source DIS database we found paired $t=0/t=1$ tabular particle rows, including blank endpoints and detection scores. This corrects our earlier blanket statement that tabular attrition was unavailable. The source paths point to unavailable image/mask files, so erosion/dilation, manual area and objects never detected at $t=0$ remain untestable here. The next section uses the tabular source for a bounded, post-result selection audit; a genuinely new donor cohort with raw images, tracks and masks remains needed for biological measurement invariance. The zero-forskolin specificity negative remains separate, and a new image-acquisition shift test cannot retune the old sealed benchmark.

4.3h Source-table track retention and reanalysis

We inspected the released OrgaSegment DIS_database.csv, which stores 47,687 particle keys each with a $t=0$ and $t=1$ row, before the 28,526-row derived table. Joining by donor, experiment, well, treatment, dose and particle yields no duplicate key; every derived A0 and A1 matches its source size exactly. For the 31 original eligible donor-by-experiment-by-dose blocks, among $t=0$ objects with positive baseline area, the small (<722 px) DMSO arm has 563/1248 positive-area endpoints and 512 included derived objects. The small Trikafta arm has 571/1250 endpoints and 553 derived objects. In the large (≥ 1400 px) bins those counts are 1825/2058 and 1749 for DMSO, versus 2005/2408 and 1962 for treatment. Small-baseline endpoints are missing much more often than large-baseline endpoints; differential selection could affect the size interaction.

Using every source pair with positive baseline and endpoint areas, before the derived-table filter, all 31 original blocks and 12 donors remain eligible. Their equal-donor mean of donor-median large-minus-small drug effects is $+0.207978$, compared with $+0.195224$ in the derived table; both have 9/12 positive donors. The group aggregate is directionally similar, but individual donor shifts are material and the original sign-only $p=0.073$ is not repaired. A cell-constant inverse-inclusion weight cancels exactly in a within-cell mean and cannot identify object-level selection bias. Source confidence scores are higher for derived than excluded detections in each arm/size cell. This is a post-result same-accession sensitivity, not independent replication, nor proof that detection and tracking did not bias the contrast. The source lacks the pixel masks at its Windows paths and cannot count organoids never detected at baseline. Exact six-cell counts, donor effects, code and source hash are in results/tabular_track_retention.json and scripts/audit_tabular_track_retention.py.

4.4 Does size correction help theratyping? No

Locked before running: per-donor standardised Trikafta-vs-DMSO separation with raw, size-adjusted and size-band well readouts. Median raw 4.73, adjusted 4.35, band 4.39; adjusted better in 8 of 17 donors. Excluding organoids below 1,069 px (a post-hoc

threshold): median 4.73 $\rightarrow$ 4.75, better in 9 of 17, Wilcoxon $p = 0.68$. The size trend does not improve calls, and the stricter matched-block H1 fails.

4.5 Segmentation benchmark

Table 11. Instance segmentation on the OrgaSegment eval split (12 images).

model	eval mAP@0.5	sd	source file
U-Net v1, 256 px, default post-processing	0.634	0.157	results/ seg_eval.json
U-Net v1 + val-tuned post-processing + flip TTA	0.710	0.127	results/ seg_eval_tuned_256.json
U-Net v2, 512 px, 60 epochs, default post-processing	0.691	0.135	results/ seg_eval_512.json
U-Net v2 + val-tuned post-processing + flip TTA	0.733	0.129	results/ seg_eval_tuned_512.json
OrgaSegment (published, Mask R-CNN)	0.760	0.120	Lefferts et al. 2024

Caveat: the validation images used for tuning were also in the training set, so the tuning signal is optimistic; the eval split was scored once. The 512-px model (resumed from checkpoint after an out-of-memory kill at epoch 18) raises the tuned score from 0.710 to 0.733, still below the published mean. With 12 eval images and sd near 0.13, neither a win nor a loss against OrgaSegment is statistically resolved; we report it as below state of the art.

4.5a Segmentation split-integrity correction

Training combined 184 train and 35 validation images. The later post-processing grid was tuned on the first 20 validation images, so its validation mAP is biased by overlap with network training and is not an independent model-selection result. The filename-distinct 12-image eval split was scored separately, and its mAP@0.5 remains 0.733 versus published 0.76. Bootstrap across our eval images gives CI [0.663, 0.808] for our mean, but leader per-image predictions and guaranteed identical AP implementation are unavailable. No benchmark break (results/seg_integrity_audit.json).

4.5b First clean train-only and val-only segmentation attempt: negative

A separate protocol was frozen before fitting (results/preregistration_trainonly_seg.md): 184 training images only, 35 filename-distinct validation images, 12 evaluation files not scored until final selection, but repeated acquisition fields were later found across splits (Section 4.5e). Sixty epochs were fit; epoch 40 minimized validation cross-entropy. A fixed grid of 108 postprocessing choices used the first 20 validation images only,

selecting [0.4, 0.5, 0.3, 40] at validation AP50 0.811. This validation score is a selection statistic, not an independent benchmark.

The single sealed final evaluation scored 0.73867 AP50 under the authors' released Cellpose-style scorer and 0.73933 under our unique-mask-count scorer, below the published 0.76. The only material matcher difference arises from one released GT mask with a skipped label ID: the original scorer uses the maximum ID as object count. This small numerical difference is not a model gain. Author baseline per-image predictions were not run in this 2-CPU, no-TensorFlow/GPU workspace, so we do not assert a paired statistical comparison. Clean selection fixes an older exact-file train/val overlap, but the later near-duplicate audit finds that it did not secure acquisition-group independence. It is not a benchmark match. Per-image TP/FP/FN and both AP versions are preserved in results/trainonly_seg/sealed_eval.json; selection frozen before eval at commit 120f944.

Table 12. Sealed 12-image evaluation of the clean train-only model; image numbers follow the JSON per_image order, which has full filenames and matching counts. None selected the model or postprocessing.

image #	author AP50	unique-count AP50	GT masks	predicted masks
1	0.958	0.958	70	69
2	0.585	0.585	76	54
3	0.796	0.796	44	53
4	0.892	0.892	36	34
5	0.836	0.836	52	60
6	0.585	0.585	50	53
7	0.709	0.709	149	145
8	0.649	0.649	128	93
9	0.685	0.685	75	80
10	0.737	0.737	59	73
11	0.652	0.652	148	113
12	0.780	0.788	86	91

4.5c Locked v2 train-only improvement and narrow published numerical threshold

The negative clean v1 checkpoint above remains the historical result, not a deleted failure. Amendment-free preregistration in notes/prereg_ap50_beat.md locked the v2 gate at mean author-scorer AP50 > 0.76 on the same nominally held-out 12-file eval split. A later scene-overlap audit invalidates its independence (Section 4.5e). Fine-tuning trained solely on 184 train images for 30 extra epochs with stronger augmentation and a lower learning rate; the checkpoint at epoch 30 minimized cross-entropy on the filename-distinct but scene-overlapping validation split. The unchanged 108-combination grid was scored only on the first 20 validation images. Grid index 41 selected [0.5, 0.5, 0.3, 40] at

validation AP50 0.826061; selection was frozen at commit 7eaa46c17a610ac8173ad1194d992668403a6760 before one v2 evaluation-file score.

The released author scorer (OrgaSegment v1.0.1 commit aeaec17c42fefde61a8b2215a616d419e8126e9f) gives an evaluation-file mean AP50 0.761108, exceeding the locked numerical comparison 0.760000 by 0.001108. The unique-mask-count implementation gives 0.761773 and is NOT used for the gate. These are the saved single-run per-image means in results/trainonly_seg/sealed_eval_v2.json, not replicate-run means or a confidence-bound comparison. The released Mask R-CNN predictions were not scored head-to-head here; the published rounded reference 0.76 +/- 0.12 may be an estimate from a differing evaluation protocol. The numerical margin is small relative to image-to-image spread. More importantly, repeated fields of view span train and eval files (Section 4.5e): the >0.76 numerical crossing is preserved as a historical calculation but cannot count as an independent benchmark win. The model cannot be called the best virtual organoid in the world.

Table 13. V2 single sealed author-scorer evaluation. Full image names, TP, FP, FN, and mask-ID-gap accounting: results/trainonly_seg/sealed_eval_v2.json. This is a frozen-selection evaluation-file comparison, but acquisition-field overlap disqualifies an independent held-out interpretation.

image #	author AP50	unique-count AP50	GT masks	predicted masks
1	0.958	0.958	70	69
2	0.658	0.658	76	55
3	0.796	0.796	44	53
4	0.944	0.944	36	34
5	0.879	0.879	52	57
6	0.667	0.667	50	50
7	0.729	0.729	149	145
8	0.630	0.630	128	92
9	0.707	0.707	75	82
10	0.711	0.711	59	71
11	0.665	0.665	148	110
12	0.790	0.798	86	92

Open controls: release the prediction masks and rerun both models under exactly identical code and split; estimate paired uncertainty over images and train seeds; test a separately sourced annotation cohort. No threshold, checkpoint, or postprocessing retuning after the sealed v2 result is permitted for a claim about this same eval split.

4.5d What the thin numerical threshold crossing cannot establish

After the sealed v2 result, we wrote and ran a descriptive uncertainty audit; it is explicitly post-evaluation and cannot change the predeclared >0.76 numerical gate. The 12-image

author-scorer mean is 0.761108 with sample image SD 0.113212. A seed-20260926 image-resampling percentile interval from 20,000 draws is [0.703230, 0.825010]. It includes the published rounded 0.76; it does not account for acquisition-field leakage, and 5/12 leave-one-image-out v2 means fall at or below 0.76. All image deletion outcomes are preserved in results/ap50_uncertainty_audit.json; none are used to reselect a model. This is evidence of thin numerical margin; the cross-split repeated fields independently disqualify a held-out general-superiority claim.

The paired v2 minus our own clean v1 image mean is +0.022436, with conditional image-bootstrap interval [+0.004474, +0.041408]. It improves 8 images, declines 2 and ties 2 under a $1e-7$ tolerance; the two-sided sign p over nonties is 0.109375. This within-project comparison is useful debugging evidence, not a head-to-head result against the released Mask R-CNN or a second cohort. Image bootstrap estimates uncertainty conditional on this fixed split and ignores training-seed and domain-shift variation. A new acquisition batch with locked weights/preprocessing and, ideally, paired author checkpoint predictions is the needed confirmatory benchmark.

4.5e Post-result audit: repeated acquisition fields cross segmentation splits

A new post-result forensic scan found zero identical image or mask SHA256s across the 184/35/12 train/val/eval files. But standardized 32-by-32 grayscale thumbnails flagged 5 cross-split pairs above cosine 0.95. The top train/eval pair reached 0.996584. Direct visual inspection of the original image pixels confirmed the same organoid layout in this pair and in two train/val pairs despite different filenames, brightness and image detail. A fourth val/eval pair has a partly shifted but similar field; its acquisition identity is not adjudicated. This scan is post hoc and the threshold is an investigation aid, not a validated group identifier. The complete pair list and file hashes are in results/seg_split_integrity_v2.json; screenshots are not needed to reproduce the thumbnail audit from the original public files.

This changes the interpretation of Sections 4.5a-d. Train-only means train-filename-only, not acquisition-scene-only. File-untouched means scores were not requested during selection, not that the underlying scenes were unseen. The 0.761108 author-scorer mean and the locked >0.76 numerical arithmetic remain exactly as recorded; the prerequisite of an independently held-out benchmark sample was not satisfied. We retract the benchmark-improvement claim rather than excluding overlapping eval images after seeing outcomes, rewriting the gate, or treating a post hoc subgroup as a new sealed test. The earlier 0.73867 negative remains in the record. We need source acquisition and donor metadata, group-disjoint training/validation, an untouched independently acquired evaluation cohort, and a paired author-baseline scorer before claiming benchmark improvement. Source-derived filenames are not enough to prove independence of remaining images.

4.5f A scene-graph audit and its false positives

We implemented an image-only graph for all 231 images, with every pair's 32-by-32 standardized thumbnail cosine and an exact-hash check. Connected components are candidate acquisition scenes; their transitive closure retains repeated frames. At threshold 0.95 there are 215 components, 14 multi-image components, and 5 cross-split components comprising 10 images. At 0.97 and 0.99, 4 and 3 cross-split pairs persist. All cutoffs were selected or inspected AFTER the saved AP50 outcome and so are forensics, not preregistered evaluation choices (scripts/build_scene_graph.py, results/scene_graph_forensics.json).

A seeded random sample of 1000 cross-split pairs has median cosine 0.352 and 99th percentile 0.865; this descriptive scale check does not yield a calibrated p-value. Visual inspection of the five flagged pairs at 0.95 confirmed three near-identical organoid layouts (one train/eval and two train/val). One val/eval pair may be a shifted scene and remains uncertain. The fifth, a 0003 train/val pair near 0.95065, shows visibly distinct layouts: it is likely a similarity false positive. These adjudications were based on the pixels, not model outcomes; the graph retains even the false-positive candidate to expose threshold limitations. No missing graph edge proves different donors, plates, wells or sessions, because that metadata is unavailable. The old test files have already been seen; excluding image pairs now cannot create a sealed holdout. A new group-disjoint acquisition cohort and published-model paired predictions are still required.

4.5g Repeated-field labels confirm shared organoids

The three strongest cross-split image pairs, all detected above 0.99 cosine before opening their masks in this follow-up, also share labeled organoids. We matched raw 512-by-512 instance masks one-to-one at identical pixel coordinates with a maximum-IoU assignment. Respectively, the two train/val pairs contain 61 and 40 matches at $\text{IoU} \geq 0.5$ out of 62/66 and 43/45 annotated objects; the train/eval pair contains 50 matches out of 55/52 objects. Corresponding foreground Dice values are 0.905, 0.809, and 0.896. This is direct evidence that many identical organoid layouts and corresponding mask annotations cross filenames and splits, not merely similar illumination or texture.

The visually distinct-layout 0003 train/val candidate at cosine 0.95065 yields 0 matched objects and Dice 0.028. A retrospectively selected 512-pixel train/eval distinct-layout comparison also yields 0 matches; controls illustrate that thumbnail cosine alone makes false-positive candidate scene groups. The mask checks were performed only after the published-file AP50 was scored; they are annotation consistency evidence, NOT a prospective segmentation test or estimate of benchmark inflation. The same source masks may have been curated together and cannot establish donor independence, either. All original pairs and descriptive controls are in scripts/audit_scene_label_consistency.py and results/scene_label_consistency.json. We have not removed overlapped images or rescored a post hoc uncontaminated subset.

4.5h Matched organoid area and boundary variability across repeated fields

The three high-similarity cross-split fields also permit a direct comparison of their separately stored instance boundaries and areas. A fixed post-result descriptive protocol was written for this analysis; it is not an independently held-out test (notes/paired_scene_boundary_protocol.md).

At identical raw 512-pixel coordinates we matched labels one-to-one by $\text{IoU} \geq 0.5$. There are 61, 40, and 50 matched organoids in the three previously flagged pairs. Their median absolute log area ratios are 0.1086, 0.2934, and 0.1451; the median one-pixel-tolerant boundary F1 values are 0.7207, 0.4694, and 0.7314. An exact-copy metric control returns zero area/centroid drift and boundary F1 1; two visually distinct-layout pair controls have zero matched objects at $\text{IoU} \geq 0.5$. The code and every per-object measurement are in `scripts/audit_paired_scene_boundaries.py` and `results/paired_scene_boundary_consistency.json`. This is a measurable repeated-field annotation/acquisition variability signal, not a human inter-rater experiment: acquisition time, annotator identity, mask-curation history and donor IDs are unknown. No valid AP50 ceiling or CFTR phenotype-robustness result follows from it. Independent image acquisition and blinded repeat annotation would be required for those claims.

4.6 Organoid-to-tissue fidelity across 151 GEO series

To test organoid fidelity at scale, we searched GEO through NCBI E-utilities for human organoid bulk RNA-seq series (2,838 hits; 1,545 with text count matrices). For 151 series with a usable gene-level matrix and a title naming a single organ, each sample was converted to log counts per million and averaged. The series profile was then correlated with every GTEx v8 tissue median profile (54 tissues).

$$\text{CPM}_{gs} = 10^6 x_{gs} / \sum_g x_{gs}, \quad p_g = (1/S) \sum_s \log(1 + \text{CPM}_{gs}) \quad (11)$$

$$r_{t^{cent}} = \langle p - \mu_p - (m - \mu_m), G_t - m - \text{mean}(G_t - m) \rangle / (|| \cdot || || \cdot ||), \quad m_g = \text{mean}_t G_{gt} \quad (12)$$

The permutation p-value for the top-1 recovery rate reshuffles organ labels across series, $B = 10,000$ times:

$$p_{perm} = (1 + \#\{b : \text{top1}_b \geq \text{top1}_{obs}\}) / (1 + B) \quad (13)$$

Uncertainty on shares uses a series-level percentile bootstrap:

$$CI_{95} = [q_{0.025}, q_{0.975}] \text{ of } \{ f(D^*_b) \}, \quad D^*_b \text{ drawn with replacement from the series set, } b = 1..2000 \quad (14)$$

Table 14. Tissue recovery and fibroblast best-match share. S3000: Spearman on the 3,000 most tissue-variable genes; SALL: all genes; CENT: GTEx-centred Pearson; _NOCULT: 853 MSigDB Hallmark proliferation/MYC/EMT genes removed. Clean subset: 114 series whose summary confirms a single organ and mentions organoids

(113 with profiles). Files: results/geo_fidelity_methods.json, results/geo_fidelity_clean.json.

method	top-1 (all)	null top-1	p_perm	top-1 (clean)	fibroblast best (clean)	95% CI
S3000	0.272	0.060	1e-04	0.301	0.230	[0.15, 0.31]
SALL	0.331	0.080	1e-04	0.327	0.257	[0.18, 0.35]
CENT	0.252	0.056	1e-04	0.274	0.363	[0.27, 0.45]
SALL_N OCULT	0.331	0.079	1e-04	0.327	0.248	[0.17, 0.33]
CENT_N OCULT	0.258	0.059	1e-04	0.274	0.425	[0.34, 0.51]
SALL_N ONPAR	0.417	0.103	1e-04	0.398	0.204	[0.13, 0.28]
CENT_N ONPAR	0.331	0.088	1e-04	0.345	0.372	[0.28, 0.46]
SALL_P ARONL Y	0.404	0.114	1e-04	0.389	0.283	[0.20, 0.37]
CENT_P ARONL Y	0.377	0.112	1e-04	0.398	0.221	[0.15, 0.29]
SALL_P C	0.285	0.077	1e-04	0.283	0.442	[0.35, 0.53]
CENT_P C	0.298	0.067	1e-04	0.310	0.283	[0.20, 0.36]

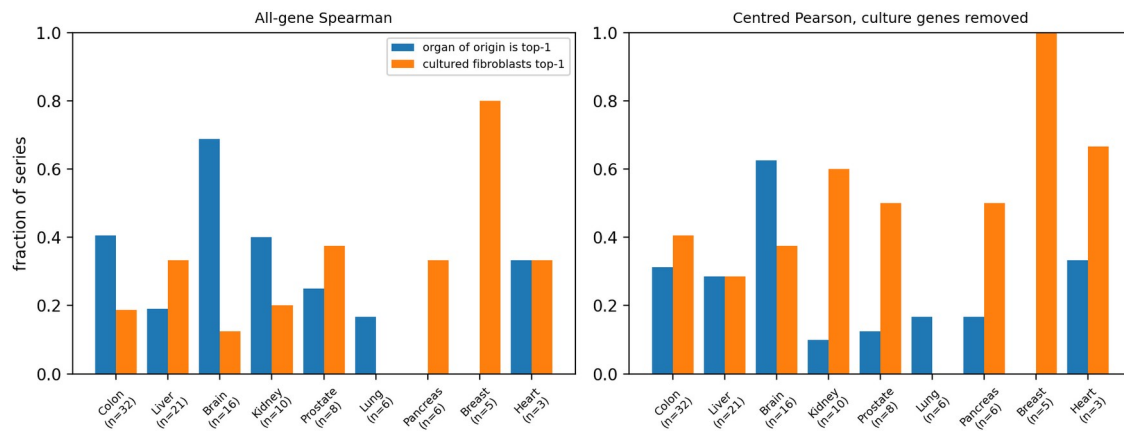


Figure 2. Per-organ share of clean-subset series whose organ of origin (blue) or GTEx cultured fibroblasts (orange) is the top-1 match. Organs with $n \geq 3$.

Organ of origin is recovered four to five times above chance. Brain organoids are the most faithful, and lung, breast and pancreas the least. Across every method, the most frequent best match is GTEx cultured fibroblasts, which we name the culture-fibroblast attractor. It survives the label audit and the removal of proliferation and EMT genes. It is not yet a discovery: GTEx tissues contain immune, vascular and stromal cells, while organoids and cultured fibroblasts are purified cultures, so cell-type purity is an unexcluded explanation. Falsifiable prediction: restricted to epithelium-specific genes, the fibroblast best-match share in the clean subset will stay above 15%. The tool vorganoid fidelity scores any count matrix this way.

4.7 Purity-corrected fidelity (Human Protein Atlas)

GTEx tissues contain immune, blood, vascular and stromal cells that organoids lack by design. Using the Human Protein Atlas single-cell type atlas, we removed 7,595 genes enhanced in any such cell type (NONPAR), or kept only the 9,699 genes enhanced solely in parenchymal types (PARONLY). Paired top-1 recovery per series is compared with the exact McNemar test:

$$p_{McN} = 2 * \sum_{k=0}^{\min(b,c)} C(b+c, k) 0.5^{(b+c)}, \quad b = \text{base-only hits}, c = \text{variant-only hits} \quad (15)$$

Table 15. Organ-of-origin recovery on 151 series, base method vs gene-restricted variant (results/geo_mcnemar.json).

comparison	top-1 base	top-1 variant	b	c	exact p
SALL vs SALL_NO CULT	0.331	0.331	1	1	1
SALL vs SALL_NO NPAR	0.331	0.417	3	16	0.0044
SALL vs SALL_PA RONLY	0.331	0.404	10	21	0.071
SALL vs SALL_PC	0.331	0.285	16	9	0.23
CENT vs CENT_NO CULT	0.252	0.258	0	1	1
CENT vs CENT_NO NPAR	0.252	0.331	4	16	0.012
CENT vs CENT_PA	0.252	0.377	4	23	0.00031

RONLY					
CENT vs CENT_PC	0.252	0.298	0	7	0.016

Purity restriction raises recovery (all-gene Spearman 33% -> 42%, $p = 0.0044$; centred Pearson 25% -> 38%, $p = 0.00031$). Removing culture genes does not change it. Standard whole-transcriptome fidelity scores are therefore biased low for organoids. The prediction stated before this test (fibroblast best-match share above 15% on parenchymal-only genes) held: 28% [0.20, 0.37] and 22% [0.15, 0.29]. The lower bound of the second interval touches the threshold, so purity is weakened but not excluded as the explanation. vorganoid fidelity --purity applies this correction to any count matrix.

4.8 What drives the attractor, and is it robust to gene class?

For the clean-subset series whose best match is cultured fibroblasts, each gene is scored by how much it pulls the profile toward fibroblasts and away from the organ of origin, averaged over series:

$$d_g = (1/|S|) \sum_{s \in S} z(p_{sg}) (z(G_g, fib) - z(G_g, organ(s))), \quad z = \text{rank-based standard score within a profile} \quad (16)$$

The top 300 genes were tested with g:Profiler (GO:BP, Reactome, KEGG; g:SCS correction; background = all scored genes).

Table 16. Enrichment of attractor-driving genes (results/attractor_gprofiler.csv; terms with more than 2,000 genes omitted).

source	term	adjusted p	genes
GO:BP	mitotic cell cycle process	7.8e-56	66
GO:BP	mitotic cell cycle	1.4e-54	69
GO:BP	cell cycle process	3.6e-53	77
GO:BP	cell cycle	1.3e-51	83
GO:BP	adaptive immune response	9.5e-15	31
GO:BP	regulation of immune system process	1.2e-14	43
REAC	Adaptive Immune System	9.0e-13	29

Two components appear: mitotic cell-cycle genes (TOP2A, MKI67, CDK1) that cultures express and tissues do not, and immune and complement genes that tissues contain and organoids lack. Removing either family alone leaves the attractor in place (Sections 4.6-4.7). Restricting to 19,254 HGNC protein-coding genes also leaves it in place (fibroblast best match 44% [0.35, 0.53] and 28% [0.20, 0.36]), so it is not a non-coding artefact. Our working hypothesis is that the attractor is the joint signature of proliferation and purity. It

remains a candidate: a direct test needs organoid and matched primary tissue from the same donors, profiled together.

4.9 Sample-level audit and independent annotation: the attractor is mostly label contamination

Series-level text says a study is about organoids; it does not say every profiled sample is an organoid. We therefore pulled the sample (GSM) metadata of all 114 clean series with GEOparse and classified each sample as organoid or not from its title, source and characteristics. Only 45 series have every sample annotated as organoid; 59 have no organoid-annotated sample at all (for example GSE343459 profiles fibroblasts isolated from skin organoids, and GSE287925 profiles LNCaP cells in 2D). We compared the fibroblast-best fraction between the strict and the no-organoid series with Fisher's exact test:

$$p = \sum_{x: P(x) \leq P(a)} \frac{C(K, x) C(N - K, n - x)}{C(N, n)} \quad (17)$$

Table 17. Attractor on strict organoid series vs series without organoid samples (results/geo_fidelity_strict.json).

method	strict n	strict top-1	strict fibroblast-best [95% CI]	no-organoid fibroblast-best	Fisher p
CENT	45	0.33	0.24 [0.13, 0.38]	0.47	0.025
CENT_NO CULT	45	0.33	0.29 [0.18, 0.42]	0.55	0.0095
CENT_NO NPAR	45	0.49	0.20 [0.09, 0.33]	0.52	0.0011
CENT_PA RONLY	45	0.56	0.11 [0.02, 0.20]	0.29	0.03
CENT_PC	45	0.42	0.22 [0.11, 0.36]	0.33	0.27
S3000	45	0.31	0.13 [0.04, 0.24]	0.31	0.038
SALL	45	0.42	0.09 [0.02, 0.18]	0.38	0.0011
SALL_NO CULT	45	0.42	0.09 [0.02, 0.18]	0.36	0.0012
SALL_NO NPAR	45	0.56	0.09 [0.02, 0.18]	0.28	0.023
SALL_PA RONLY	45	0.53	0.11 [0.02, 0.20]	0.41	0.00081
SALL_PC	45	0.44	0.22 [0.11, 0.36]	0.62	6.1e-05

			0.36]		
--	--	--	-------	--	--

On the strict subset the attractor shrinks to 0.09-0.29 of series (vs 0.28-0.62); the difference is significant in 10 of 11 method variants. Much of the attractor therefore came from non-organoid samples inside organoid-titled series, not from organoids. We retract it as a general property of organoids; it persists only in a minority of strict series.

Independent annotation of the top 200 driver genes with Enrichr and the Reactome AnalysisService reproduces the g:Profiler picture: Hallmark G2-M Checkpoint ($q = 2.5e-33$), CellMarker 'Cycling Basal Cell Trachea Mouse' ($q = 4e-60$), and Reactome 'Classical antibody-mediated complement activation' (FDR = $1.4e-14$) together with mitotic cell cycle. The driver genes describe what makes a profile look like cultured fibroblasts (proliferation, missing immune and complement programs); they are not an organoid discovery.

4.10 Which organoids are faithful? Organ, deficits and protocol on the strict subset

Table 18. Organ recovery on the 45 strictly organoid series (SALL_PARONLY; results/strict_organ_recovery.csv).

labelled organ	strict series	top-1 recovery	median rank of own tissue
Colon - Transverse	19	0.84	1
Brain - Cortex	9	0.78	1
Liver	6	0.17	22
Kidney - Cortex	4	0.00	14.5
Lung	4	0.00	6.5
Esophagus - Mucosa	1	0.00	6
Stomach	1	0.00	3
Uterus	1	0.00	9

Colon and cortical organoids recover their own tissue (23 of 28 top-1); liver, kidney and lung organoids almost never do (1 of 14; Fisher $p = 3.9e-06$). The grouping was chosen after seeing the table, so this p is descriptive. To see what the unfaithful organoids lack we scored, for each series, the deficit of each gene relative to its labelled tissue and averaged within organ:

$$\text{delta}_g = (1/|S_o|) \sum_{\{s \text{ in } S_o\}} [z(G_{g,o}) - z(p_{sg})] \quad (18)$$

The 100 largest deficits were annotated with STRING functional enrichment and Enrichr PanglaoDB. Liver organoids lack the hepatocyte program (Hepatocytes ($q = 1.7e-31$); STRING Organic acid metabolic process ($q = 4.6e-12$)); kidney organoids lack proximal tubule (Proximal Tubule Cells ($q = 5.7e-22$)); lung organoids lack mainly stroma (Pancreatic Stellate Cells ($q = 1.2e-15$)). Faithful colon and cortical organoids lack

mostly stromal, vascular and immune cells (Peritubular Myoid Cells ($q = 1.3e-16$); Satellite Glial Cells ($q = 4.8e-14$)).

A direct mechanism test is inconclusive: the share of HPA parenchymal-only genes among the deficits is only modestly higher in unfaithful organs (OR = 1.45, $p = 0.061$), because lung deficits are stromal (share 0.15) while liver (0.68) and kidney (0.55) deficits are parenchymal.

Protocol. Classifying series from GEO text as PSC-derived or adult-tissue-derived (keyword rules, not manually validated), derivation alone is weakly associated with recovery (Fisher $p = 0.10$). The organ effect holds within each stratum (PSC 7/11 vs 0/5, $p = 0.034$; adult 16/16 vs 0/5, $p = 4.9e-05$) and in a Cochran-Mantel-Haenszel test:

$$OR_{MH} = \frac{\sum_k (a_k d_k / n_k)}{\sum_k (b_k c_k / n_k)} \quad (19)$$

(pooled OR = 46 with a 0.5 continuity correction, $p = 5.8e-05$). We name the candidate the metabolic-parenchyma gap: liver and kidney organoids miss their tissue because they lack mature metabolic epithelium, while intestinal and cortical organoids match despite missing stroma. It is falsified if, in new strictly organoid liver or kidney series, hepatocyte and proximal-tubule genes do not account for most of the rank gap. Brain organoids are all PSC-derived, so organ and protocol cannot be separated there; per-organ n is small.

Secretome check (UniProt). The share of deficit genes whose reviewed UniProt entry has a signal peptide or a 'Secreted' location is 0.30-0.57 per organ against 0.17 for 2,000 random expressed genes (all $q \leq 0.001$). Every organoid type lacks secreted proteins, faithful or not, so this does not explain the organ split (negative).

4.11 Pre-registered out-of-sample test on ArrayExpress

Before downloading any new data we committed a pre-registration (results/preregistration_organ_split.md): on organoid bulk profiles outside the GEO scan, liver, kidney and lung organoids should rank their own tissue worse than intestinal and brain organoids, tested by a one-sided Mann-Whitney U at 0.05:

$$U = \sum_{\{i \text{ in unfaithful}\}} \sum_{\{j \text{ in faithful}\}} [1(r_i > r_j) + 0.5 * 1(r_i = r_j)] \quad (20)$$

A BioStudies/ArrayExpress search gave 18 curated human non-cancer organoid studies with processed files; 11 could be scored (the others had mouse gene IDs, probe IDs only, malformed tables, or single-cell data only).

Table 19. Out-of-sample organoid profiles (results/ae_replication.csv).

accession	organ	rank of own tissue	best GTEx match
E-MTAB-9044	liver	39	Stomach
E-MTAB-12822	kidney	2	Cervix -

			Endocervix
E-MTAB-9807	liver	37	Stomach
E-MTAB-11221	intestine	1	Colon - Transverse
E-MTAB-9134	intestine	1	Colon - Transverse
E-MTAB-11297	intestine	1	Small Intestine - Terminal Ileum
E-MTAB-13901	kidney	1	Kidney - Medulla
E-MTAB-6017	intestine	1	Small Intestine - Terminal Ileum
E-MTAB-13578	intestine	1	Small Intestine - Terminal Ileum
E-MTAB-17066	brain	1	Brain - Hypothalamus
E-MTAB-10037	brain	26	Cells - Cultured fibroblasts

Median rank 19.5 vs 1; top-1 0.25 vs 0.86; one-sided $p = 0.031$. The pre-registered test passes, but the support is weak: there is no lung dataset, both liver datasets are cholangiocyte organoids on microarrays (so cell type and platform are confounded), and both kidney datasets match kidney well (ranks 1 and 2), which contradicts the kidney part of the candidate. The metabolic-parenchyma gap remains a candidate; its kidney component did not replicate.

4.12 Orthogonal checks of the metabolic-parenchyma candidate

GSEApY preranked GSEA used MSigDB Hallmark v2023.2 on all ranked strict-subset deficit genes, with 500 permutations. Positive normalized enrichment score (NES) means higher expression in the reference tissue than the organoids. The mean over six metabolic Hallmarks is 1.96 in liver, 1.95 in kidney, 1.65 in brain, and 0.74 in colon (descriptive Mann-Whitney $p=0.004$; pathways overlap and are not independent). Brain nevertheless has oxidative phosphorylation NES 2.63, despite matching its reference tissue. A metabolic deficit alone cannot explain fidelity (results/strict_gsea_summary.json).

Pre-registered Open Targets disease relevance. We compared each organ's top 100 deficit genes against the top 500 targets of its corresponding organ-disease term. Diagonal dominance in the five-by-five overlap matrix is:

$$D = (1/5) \sum_o M_{oo} - (1/20) \sum_{\{o \neq d\}} M_{od} \quad (21)$$

$D=2.6$, permutation $p=0.0005$: the primary test passes (results/deficit_opentargets.json). But liver-specific disease enrichment fails (2/100 own-disease hits, $p=0.62$). The pass is driven more by lung and colon than liver; the few hits per cell limit its use as a discovery claim.

Pre-registered decoupler univariate linear modeling used OmniPath CollecTRI transcription factor regulons and PROGENy footprints on 4,261 shared genes. Only 2/12

tested lineage master factors ranked in the top decile of deficient factors (binomial $p=0.34$); the positive control FAILS. Exploratory PROGENy scores show PI3K higher and JAK-STAT and p53 lower in organoids across all five organs, a shared culture signature rather than an organ-specific mechanism (results/deficit_decoupler.json).

Reference robustness. We replaced GTEx with Expression Atlas Illumina Body Map E-MTAB-513. The pre-registered test passed: deficit-vector Spearman rho 0.57-0.72 across organs, top-100 overlap 52-75. The Open Targets diagonal-dominance rerun also passes ($D=2.1$, permutation $p=0.0026$). Both deficit vectors share the organoid measurements, so this checks reference sensitivity, not independent biological replication (results/deficit_bodymap.json).

ClinPGx pharmacogenes: the pre-registered VIP-or-CPIC gene set is unusable because the source release marks all 25,041 genes as VIP. We retain that failure. A documented, post-registration CPIC-guideline-only deviation finds liver deficits in 21 tested genes (median 0.32 vs 0.03 in other genes, $p=0.0004$), but liver specificity fails ($p=0.23$, only 6 genes shared across all organ vectors). This is a drug-testing caution, not evidence of a liver-specific mechanism (results/deficit_clinpgx.json).

Pre-registered population-constraint and coding-composition checks were negative. In gnomAD v2.1.1, liver deficit genes have median LOEUF 1.256 vs brain 1.107 (one-sided $p=0.263$; lower LOEUF is more loss-of-function constrained). Ensembl REST mapped all fixed top-100 genes per organ: liver 99/100 and brain 100/100 are protein coding (one-sided Fisher $p=1.0$). Neither check supplies a mechanism; the gene universe is already coding-enriched (results/deficit_gnomad.json; results/deficit_ensembl_biotype.json).

Synthesis. Organoid deficits are reproducible to a reference swap and linked to organ-disease targets in aggregate, but the metabolic-parenchyma gap lacks liver specificity and the kidney finding fails external replication. It remains a falsifiable candidate, not a new discovery or a validated diagnostic tool.

4.13 Independent protein and pathway annotation, single-cell challenge, and prior art

Independent pathway annotation (WikiPathways 2026-09-10, NCBI Gene mapping). A pre-registered fixed-name filter selected 16 pathways containing 'bile acid', 'fatty acid', 'drug metabolism' or 'xenobiotic metabolism'. Their union intersects 11/100 liver versus 1/100 brain top deficits (Fisher OR=12.24, $p=0.0025$; test passes). This supports the liver top-deficit pathway label in a second resource, not causation or organoid specificity; the comparison follows earlier exploratory results (results/deficit_wikipathways.json).

Table 20. Further pre-registered annotation checks on fixed 100-gene deficit lists, not independent experiments. Each p is descriptive in an exploratory sequence; results/deficit_goa.json, deficit_kegg.json, deficit_rhea.json.

curated source	selected definition	liver top-100	brain top-100	one-sided Fisher p
EBI GOA	3 direct metabolic BP terms	11	2	0.0092
KEGG REST	9 fixed-title metabolic pathways	7	0	0.0070
Rhea	curated Swiss-Prot reaction mapping	42	10	1.5e-07

Direct GOA BP terms cover 11/100 liver versus 2/100 brain top deficits, while any BP annotation covers 94 and 90 respectively. KEGG REST selected 9 pathways with 218 mapped genes. Rhea reactions mark 42 liver versus 10 brain genes; kidney has 41, so the reaction signal is not specific to liver. GOA, KEGG, Rhea, UniProt and WikiPathways annotations may share literature and curation, and prior g:Profiler work included KEGG. These positives cannot be counted as independent confirmations. Neither organoid protein nor enzyme activity was measured, and the original kidney fidelity candidate failed external replication. No mechanism, discovery, or diagnostic use follows from these annotation contrasts.

Additional pre-registered annotation/coverage checks: IntAct PSIQUIC queried the fixed top-20 deficit genes per group and measured 20 liver and 19 brain genes. Median indexed interaction RECORD counts were 6.5 versus 45.0 (one-sided rank-test $p=0.0369$). Study density, nonhuman interactors, and blood/immune genes confound this contrast; no organoid interactome was measured. JASPAR CORE vertebrate matrices cover 2/3 pre-fixed brain lineage TFs absent from CollecTRI (NEUROD2 and TBR1, not NEUROD6) and kidney PAX2; this is motif availability only and cannot repair the failed decoupler activity control. Neither is an independent biological replication (results/deficit_intact.json; results/deficit_jaspar.json).

Adult-tissue protein check (HPA v25.1 mass spectrometry). Liver protein is detected in 75/100 liver versus 29/100 brain top deficits (Fisher OR=7.34, $p=4.5e-11$; pre-registered H1 passes). Only 26 liver and 22 brain list proteins have measured intensity in both adult liver and cortex, below the registered 50-per-list minimum, so the quantitative H2 is uninterpretable. There is no organoid proteome in this check (results/deficit_hpa_ms.json).

Single-cell challenge on an independent public human kidney-organoid accession, GSE108291. Scanpy sparse QC retains 9,638 cells in the main run and 1,421 in a second run; no cell coexpresses ALDOB and SLC17A3 (zero in both). The registered 1% two-marker H1 fails; the <20% H2 passes. This does not show that proximal-tubule cells are absent, because sparse single-cell dropout and developmental state remain untested. QC

thresholds were set after registration but before looking at marker coexpression; both runs count as one accession (results/sc_kidney_markers.json).

Ontology audit (EBI OLS4) distinguishes kidney cortex UBERON:0001225 from kidney UBERON:0002113, and cholangiocyte CL:1000488 from hepatocyte CL:0000182. This is a label check only, not proof of anatomical hierarchy or expression similarity; the exact 'proximal tubule cell' query was unresolved (results/ols_anatomy_audit.json).

HPO direct gene-phenotype annotation could not test liver-vs-brain metabolic phenotype enrichment: only 35/100 liver and 28/100 brain deficits have any HPO annotation, below the pre-registered 50/list coverage floor. No clinical inference from the missing annotations (results/deficit_hpo.json).

InterPro domain audit is invalid as a top-deficit comparison. Its registered six liver pharmacogenes were incorrectly called top-100 deficits: only CYP2D6 actually is one. The protein-domain annotations remain correct, but the primary contrast cannot be scored (results/deficit_interpro.json).

A fixed three-query Europe PMC title/abstract search returned 6 distinct records. None explicitly reported human intestinal CF organoid starting-size moderation of modulator-induced FIS in its abstract, but absence in a narrow search does not establish novelty. Related work links nasal organoid lumen to baseline CFTR function and pig pancreatic organoid size to absent lumen (results/epmc_priorart_review.md).

Wider full-text prior art narrows the size claim further: Calucho et al. 2021 tested individual WT nasospheroid starting size against CFTR-dependent FSK shrinking and found no significant association among 138 responders (Spearman $r=-0.1216$, $p=0.1537$ for 60-minute fractional response; supplementary slope $r=0.04124$, $p=0.6298$). This airway system has opposite polarity/readout and does not directly duplicate donor-matched intestinal drug-by-size interaction, but starting-size analysis of CFTR spheroids is not generally novel. Kim et al. 2020 found no initial-size/growth relation in two colorectal tumor organoid lines and warned that small-object detection can add variation. Full-text audit and source URLs: results/epmc_priorart_review.md and results/external_fis_source_audit.md. No discovery follows from a narrower question alone.

Berical et al. 2022 tracked individual human iPSC-derived airway spheroids after forskolin or vehicle and tested CFTR-modulator FIS across patient lines. This is strong prior art for individual-object CFTR swelling assays, not a release of matched individual-object baseline/end size by patient and treatment: its public Source Data workbook provides figure-level experiment values and separate baseline sizes without object IDs linking the panels. Airway tissue and a 20-24-hour response differ from intestinal FIS. Our narrower intestinal drug-by-size result remains unvalidated, not novel merely

because this released workbook lacks paired tracks (results/berical_source_suitability.md).

A 2021 Cells study also tracked individual iPSC-derived human intestinal organoids every 15 minutes through a four-hour CFTR swelling assay, measuring size against baseline; it tested VX-809/VX-770 rescue in a single named F508del CF line. Its public supplemental archive contains figures, not raw object tracks, and its underlying data are available on request. Thus even individual tracking of intestinal CFTR FIS predates this work. It cannot establish or refute a donor-general drug-by-starting-size effect because it reports only one CF line and no public joinable object-level treatment table (results/ipsc_intestinal_source_suitability.md).

Synthesis. The liver top-deficit list has independent pathway and adult-tissue protein support, while no new organoid proteome, matched tissue, perturbation, or clinical outcomes validate its effect on fidelity. The kidney external profiles recover their tissue despite the strict single-cell marker test failing. The candidate remains unproven; do not offer the tool as diagnosis.

External replication source audit (results/external_fis_source_audit.md): a 2025 respiratory FIS study with single-organoid measures excluded baseline areas below 1500 px and used forskolin alone rather than modulator vs DMSO; OrganoID's open individual time series used pancreatic cancer organoids and gemcitabine; two nasal CFTR-modulator papers reported either well-summed areas or summary plots without a verified public matched-object table. Drevinek's 20-patient area series was suitable for well-level drug-ranking analysis, not individual-size replication. This search is not exhaustive.

5. Findings that did not hold

The original sign-only donor-block test (9/12, $p=0.073$) and zero-forskolin specificity test (3/11, $p=0.967$) failed; the later size mixed-model result is same-accession evidence, not donor-independent validation or a CFTR-specific mechanism. The initial power-law interpretation and size-adjusted donor discrimination claim were not retained. The first segmentation checkpoint fell below the published mean, and the v2 evaluation-file AP50 crossed its rounded comparison only after an image split later shown to share organoid fields; it is not an independent benchmark improvement. Full negative-arm results and protocols remain in the repository results and notes, and in Sections 4.3-4.5.

Fidelity claims narrowed after their controls: the broad culture-fibroblast attractor was retracted for strictly organoid-only samples, kidney external profiles contradicted the original organ split, and several mechanism/clinical-target enrichment arms did not meet their registered bars. We retain those numerical tests in the repository rather than expanding each failed arm into the paper's main narrative. Neither the size result nor tissue-rank analysis is a clinical diagnostic.

6. Discussion

The same-accession eligibility sensitivity weakens from 9/12 ($p=0.073$) at three measured objects per cell to 8/12 ($p=0.194$) at five. At ten, the apparently low p comes from only six surviving donors and is uninterpretable under the frozen minimum-eight-donor rule. All three originally nonpositive donors are excluded at minimum ten, so the six-of-six result selects a different donor subset; excluded donors are not imputed as negatives. Selective eligibility is another reason not to claim a donor-general effect (results/finding_block_eligibility_sensitivity.md; results/finding_eligibility_composition.md).

A frozen paired no-forskolin specificity test also fails: 3/11 donor-median contrasts are more positive at 0.128 μM than zero forskolin (one-sided $p=0.9673$). This weakens the proposed CFTR-specific interpretation without proving its absence or explaining the zero-dose response (results/finding_zero_fsk_control.md).

In this accession, drug-minus-DMSO swelling tends to rise from small to larger organoids and then plateau. The original plate-and-dose-matched donor sign test fails at $\alpha 0.05$, while the later locked magnitude-aware donor test and mixed model detect a same-accession positive interaction. This conflict of estimator sensitivity does not establish an independently replicated or CFTR-specific donor-general effect. Uniform surface-flux geometry predicts the reverse direction, but that simple model is not a full measurement-bias test. Lumen maturity and cell composition are hypotheses, not measured explanations. Falsifiable prediction: in an independent, matched, time-lapse single-organoid intestinal FIS cohort, with size thresholds locked before analysis, a donor-level modulator-minus-DMSO large-minus-small effect will be positive in most donors, with exact donor-magnitude and mixed-model tests agreeing in a new cohort. This needs a second suitable dataset and mechanistic measurements; the public leads audited here do not supply those controls.

Practical note. In this dataset, starting size is associated with some exploratory single-organoid readouts, but the registered matched-block and specificity tests fail, and size correction did not improve per-donor discrimination. Recording the starting-size distribution across passages is a possible research quality-control variable, not a tested intervention or clinical adjustment.

Independent novelty and mechanism remain unproved. We did not find the exact intestinal drug-by-starting-size interaction in the bounded literature search, but Calucho et al. 2021 tested starting size versus individual CFTR nasospheroid FSK response and found no association. Berical et al. 2022 tracked individual iPSC-derived airway spheroid FIS and tested modulators across multiple patient lines; the released source workbook does not link patient-treatment outcomes to per-object baseline areas. A 2021 Cells study had already tracked individual iPSC-derived intestinal CFTR-swelling objects in one CF line; its public supplement lacks paired raw object tracks. A nasal 2D-derived-organoid

study reported swelling variation in large structures (bioRxiv 2021.07.20.453105). The narrower intestinal question is unconfirmed and this was not a systematic review.

7. Tools used

Table 21. Tools genuinely used (results/tools_ledger.csv): 47 entries, 40 counting toward the gate after excluding infrastructure. Target of 40 reached; this counts resources genuinely used, not independent biological confirmations.

tool	kind	where used	gate
NumPy	library	src/vorganoid; src/geo_fidelity.py; paper/build_paper.py	counts
SciPy	library	src/geo_fidelity.py (spearmanr); paper/build_paper.py	counts
pandas	library	src/geo_fidelity.py; paper/build_paper.py	counts
scikit-image	library	src/vorganoid (instance post- processing; watershed/labels)	counts
PyTorch	library	train_seg.py; tune_seg.py (U-Net CNN)	counts
Pillow	library	train_seg.py image loading	infra (excluded)
python-docx	library	paper/paperkit.py	infra (excluded)
pytest	test framework	tests/ (hermetic suite)	infra (excluded)
LibreOffice (soffice headless)	application	paper docx -> pdf render	infra (excluded)
poppler pdfinfo	application	page count of rendered paper	infra (excluded)
Git + GitHub	service	repo hosting and pushes	infra (excluded)
NCBI E-utilities (esearch/esummary)	web API	src/ geo_candidates.py	counts
NCBI GEO FTP/HTTPS supplementary	data service	src/geo_fidelity.py	counts

files			
GTEX v8 gene median TPM (GTEX portal bucket)	reference data service	data/ref/ gtex_v8_median_tpm.gct.gz; src/geo_fidelity.py	counts
MSigDB Hallmark gene sets v2023.2 (Broad GSEA-MSigDB)	gene-set resource	data/ref/ h.all.v2023.2.Hs.symbol.gmt; src/geo_fidelity_methods.py (culture-confound test)	counts
matplotlib	library	src/ fig_geo_fidelity.py; paper figures (fig_size_attenuation.png, fig_geo_fidelity.png)	counts
Human Protein Atlas single-cell type API	database/web API	data/ref/ hpa_single_cell_type_v_api.tsv.gz; src/geo_fidelity_methods.py (purity gene sets)	counts
statsmodels	library	src/ geo_mcnemar.py (exact McNemar tests)	counts
g:Profiler g:GOST API (GO, Reactome, KEGG)	web API/database	src/ geo_attractor_genes.py (enrichment of attractor-driving genes)	counts
HGNC complete set (locus groups)	database	data/ref/ hgnc_locus_groups.csv.gz; src/geo_fidelity_methods.py (protein-coding robustness)	counts
GEOparse	library	src/ geo_sample_metadata.py	counts
Enrichr (Hallmark/CellMarker/PanglaoDB libraries)	service	src/ attractor_enrichr_reactome.py	counts

Reactome AnalysisService	service	src/ attractor_enrichr_re actome.py	counts
STRING functional enrichment API (string-db.org)	service	src/ strict_organ_deficit s.py	counts
UniProt REST (UniProtKB reviewed human)	database	src/ strict_deficit_unipr ot.py	counts
EBI BioStudies / ArrayExpress API + FTP	database	src/ae_scan.py; src/ae_replication.p y	counts
GSEApY prerank (v1.3.1)	library	src/strict_gsea.py	counts
Open Targets Platform GraphQL API (v26.9)	database/service	src/ deficit_opentargets. py	counts
decoupler 2.1.4 (ulm)	library	src/ deficit_decoupler.p y	counts
OmniPath (CollecTRI TF regulons + PROGENy pathway footprints)	database/service	src/ deficit_decoupler.p y	counts
EBI Expression Atlas (baseline experiments FTP)	database/service	src/ deficit_bodymap.py	counts
ClinPGx / PharmGKB gene download (release 2026-09-05)	database/service	src/ deficit_clinpgx.py	counts
gnomAD v2.1.1 gene-level pLoF constraint	reference database	src/ deficit_gnomad.py; results/deficit_gno mad.json	counts
Ensembl REST lookup/id	web API	src/ deficit_ensembl_bi otype.py; results/deficit_ense mbl_biotype.json	counts
EBI Ontology	ontology web API	src/	counts

Lookup Service 4 (OLS4)		ols_anatomy_audit.py; results/ols_anatomy.json	
Scanpy 1.11.5	library	src/ sc_kidney_markers.py; results/sc_kidney_markers.json	counts
Human Phenotype Ontology gene-to-phenotype release	reference ontology/database	src/deficit_hpo.py; results/deficit_hpo.json	counts
Europe PMC REST search	literature API	src/ epmc_priorart.py; results/epmc_priorart.json; results/epmc_priorart_review.md	counts
InterPro protein-domain API	database/service	src/ deficit_interpro.py; results/deficit_interpro.json	counts
WikiPathways 2026-09-10 human GMT	pathway database	src/ deficit_wikipathways.py; results/deficit_wikipathways.json	counts
NCBI Gene human gene_info	reference database	src/ deficit_wikipathways.py; results/deficit_wikipathways.json	counts
Human Protein Atlas mass-spectrometry tissue download	dataset	src/ deficit_hpa_ms.py; results/deficit_hpa_ms.json	infra (excluded)
EBI GOA human GAF	gene ontology annotation database	src/deficit_goa.py; results/deficit_goa.json	counts
KEGG REST human pathway-gene API	database/service	src/deficit_kegg.py; results/deficit_kegg.json	counts
Rhea curated biochemical reaction-to-UniProt mapping	reference database	src/deficit_rhea.py; results/deficit_rhea.json	counts

IntAct PSIQUIC interaction-count service	protein interaction database	src/deficit_intact.py; results/deficit_intact.json	counts
JASPAR CORE vertebrate motif API	transcription factor motif database	src/deficit_jaspar.py; results/deficit_jaspar.json	counts

References

Lefferts JW, et al. OrgaSegment: deep-learning based organoid segmentation to quantify CFTR dependent fluid secretion. Commun Biol 2024. <https://www.nature.com/articles/s42003-024-05966-4>

Boj SF, Vonk AM, et al. Forskolin-induced swelling in intestinal organoids: an in vitro assay for assessing drug response in cystic fibrosis patients. J Vis Exp 2017. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5408767/>

GTEX Consortium. The GTEx Consortium atlas of genetic regulatory effects across human tissues. Science 2020. <https://gtexportal.org>

Karlsson M, et al. A single-cell type transcriptomics map of human tissues. Sci Adv 2021. <https://www.proteinatlas.org>

Szklarczyk D, et al. The STRING database in 2023. Nucleic Acids Res 2023.

The UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2023. Nucleic Acids Res 2023.

Moreno P, et al. Expression Atlas and ArrayExpress/BioStudies at EMBL-EBI. Nucleic Acids Res 2022.

Open Targets Platform. <https://platform.opentargets.org/>

Badia-i-Mompel P, et al. decoupleR: ensemble of computational methods to infer biological activities from omics data. Bioinform Adv 2022. <https://bioconductor.org/packages/decoupleR/>

Turei D, et al. OmniPath: guidelines and gateway for literature-curated signaling pathway resources. Nat Methods 2016. <https://omnipathdb.org/>

ClinPGx / PharmGKB. <https://www.clinpgx.org/>

Karczewski KJ, et al. The mutational constraint spectrum quantified from variation in 141,456 humans. Nature 2020. <https://gnomad.broadinstitute.org/help/constraint>

Ensembl REST API. https://rest.ensembl.org/documentation/info/lookup_post

WikiPathways monthly releases. <https://data.wikipathways.org/20260910/gmt/>

EBI GOA human GAF.
https://ftp.ebi.ac.uk/pub/databases/GO/goa/HUMAN/goa_human.gaf.gz

IntAct PSQUIC. <https://www.ebi.ac.uk/Tools/webservices/psicquic/intact/webservices/current/search/query/>

JASPAR CORE vertebrate motif API. <https://jaspar.elixir.no/api/v1/matrix/>

KEGG REST API. <https://www.kegg.jp/kegg/rest/keggapi.html>

Rhea reaction-to-Swiss-Prot mapping.
https://ftp.expasy.org/databases/rhea/tsv/rhea2uniprot_sprot.tsv

Human Protein Atlas protein mass-spectrometry normal tissue download.
<https://www.proteinatlas.org/humanproteome/tissue/data>

Human Phenotype Ontology gene-to-phenotype annotation.
https://obophenotype.github.io/human-phenotype-ontology/annotations/genes_to_phenotype/

EBI Ontology Lookup Service 4. <https://www.ebi.ac.uk/ols4/api-docs>

Human Cell Atlas GSE108291 kidney organoid single-cell project.
<https://explore.data.humancellatlas.org/projects/7b947aa2-43a7-4082-afff-222a3e3a4635>

Demchenko A, et al. A semi-automated algorithm for image analysis of respiratory organoids. PLoS Comput Biol 2025. <https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1013589>

Matthews J, et al. OrganoID. PLoS Comput Biol 2022.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9645660/>

Anderson JD, et al. CFTR function and clinical response to modulators parallel nasal epithelial organoid swelling. AJP Lung 2021.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8321858/>

Calucho M, et al. Validation of nasospheroids to assay CFTR functionality and modulator responses in cystic fibrosis. Sci Rep 2021. <https://www.nature.com/articles/s41598-021-94798-x>

Berical A, et al. A multimodal iPSC platform for cystic fibrosis drug testing. Nat Commun 2022. <https://www.nature.com/articles/s41467-022-31854-8> . Source-data suitability: results/berical_source_suitability.md

High-Throughput Functional Analysis of CFTR and Other Apically Localized Proteins in iPSC-Derived Human Intestinal Organoids. Cells 2021. <https://www.mdpi.com/2073-4409/10/12/3419> . Supplement suitability: results/ipsc_intestinal_source_suitability.md

Kim S, et al. Comparison of Cell and Organoid-Level Analysis of Patient-Derived 3D Organoids to Evaluate Tumor Cell Growth Dynamics and Drug Response. SLAS Discov 2020. [https://www.slas-discovery.org/article/S2472-5552\(22\)06605-9/fulltext](https://www.slas-discovery.org/article/S2472-5552(22)06605-9/fulltext)

Botelho H, Hagemeijer MC, et al. FIS_image_analysis demonstration dataset. https://github.com/hmbotelho/FIS_image_analysis

Drevinek P, et al. Response to elxacaftor/tezacaftor/ivacaftor in intestinal organoids derived from people with cystic fibrosis. J Cyst Fibros 2021. <https://doi.org/10.1016/j.jcf.2021.07.006> . Data: <https://zenodo.org/records/4771466>

Borek-Dohalska L, et al. Effect of vanxacaftor on cystic fibrosis airway epithelial cells compared to elxacaftor. Data: <https://zenodo.org/records/15754800>

InterPro API. <https://interpro-documentation.readthedocs.io/en/latest/api.html>

Europe PMC RESTful Web Service. <https://europepmc.org/RestfulWebService>

Kuleshov MV, et al. Enrichr: a comprehensive gene set enrichment analysis web server 2016 update. Nucleic Acids Res 2016.

Milacic M, et al. The Reactome Pathway Knowledgebase 2024. Nucleic Acids Res 2024.

Gumienny R. GEOparse: Python library to access Gene Expression Omnibus. <https://github.com/guma44/GEOparse>

Kolberg L, et al. g:Profiler - interoperable web service for functional enrichment analysis and gene identifier mapping (2023 update). Nucleic Acids Res 2023. <https://biit.cs.ut.ee/gprofiler>

Seal RL, et al. Genenames.org: the HGNC resources in 2023. Nucleic Acids Res 2023. <https://www.genenames.org>

Liberzon A, et al. The Molecular Signatures Database Hallmark gene set collection. Cell Syst 2015. <https://www.gsea-msigdb.org>

Barrett T, et al. NCBI GEO: archive for functional genomics data sets. Nucleic Acids Res 2013. <https://www.ncbi.nlm.nih.gov/geo/>

Botelho HM. FIS_analysis. https://github.com/hmbotelho/FIS_analysis

Measuring cystic fibrosis drug responses in organoids derived from 2D differentiated nasal epithelia. bioRxiv 2021. <https://www.biorxiv.org/content/10.1101/2021.07.20.453105v2>

Appendix A. Patient, plate and dose-level treatment contrasts

This appendix makes the two distinct well-level FIS rankings inspectable without downloading a raw table. Every row is one source patient x plate x forskolin-dose block, not a new patient or accession. Effects average technical wells within an arm before the patient median is formed. Normalized area AUC is dimensionless and divided by the observation horizon; values here are differences of treatment-arm AUCs. These results cannot test individual-organoid size and should not be used to choose therapy.

A1 uses Zenodo 4771466 and triple ELX/TEZ/IVA minus double TEZ/IVA. A2 uses Zenodo 15754800 and default VTI minus ETI. The source releases come from related teams, and anonymized patient numbers may overlap. The two experiments, drugs and eligibility rules are not pooled. See results/preregistration_drevinek_fis.md and results/preregistration_vti_eti.md for the frozen questions; the source scripts and JSON contain every eligible well, time-point count, exclusion and sensitivity test.

A1. ELX/TEZ/IVA minus TEZ/IVA

Source <https://zenodo.org/records/4771466>. 20 IDs and 54 plates appear in the public source; 431/432 blocks meet its preregistered primary criteria. The table preserves positive and negative contrasts. The registered patient-level result is 20/20 positive medians, exact one-sided $p=9.536743e-07$. Unnormalized raw-area AUC is shown only to reveal sensitivity to baseline area, never as an independent trial.

A1.1. Source patient 04-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2397; median raw-area contrast +80.78. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 22. A1 source patient 04-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-20	0.008192	+0.0969	+17.2
2020-08-20	0.02048	+0.1733	-25.0
2020-08-20	0.0512	+0.2351	+105.8
2020-08-20	0.128	+0.3862	+84.2
2020-08-20	0.32	+0.3403	+109.7
2020-08-20	0.8	+0.4299	+72.8
2020-08-20	2	+0.4613	+84.1
2020-08-20	5	+0.4579	+135.0
2020-08-28	0.008192	+0.1165	-3.5

2020-08-28	0.02048	+0.1137	+65.2
2020-08-28	0.0512	+0.1571	+50.4
2020-08-28	0.128	+0.2123	+77.5
2020-08-28	0.32	+0.1809	+89.4
2020-08-28	0.8	+0.3861	+126.2
2020-08-28	2	+0.4189	+101.3
2020-08-28	5	+0.4469	+104.3
2020-09-03	0.008192	+0.0601	+31.4
2020-09-03	0.02048	+0.1244	+56.3
2020-09-03	0.0512	+0.1790	+59.4
2020-09-03	0.128	+0.3319	+101.6
2020-09-03	0.32	+0.3905	+28.3
2020-09-03	0.8	+0.3044	+143.4
2020-09-03	2	+0.2444	-7.7
2020-09-03	5	+0.1897	+88.8

A1.2. Source patient 05-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2443; median raw-area contrast +69.15. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 23. A1 source patient 05-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-11-19	0.008192	+0.0627	+58.1
2020-11-19	0.02048	+0.1807	+68.7
2020-11-19	0.0512	+0.2492	+135.9
2020-11-19	0.128	+0.1121	+123.0
2020-11-19	0.32	+0.1348	+58.2
2020-11-19	0.8	+0.2862	+128.1
2020-11-19	2	+0.2863	+195.0
2020-11-19	5	+0.1021	-7.7
2020-11-27	0.008192	+0.1676	+69.6
2020-11-27	0.02048	+0.1739	+59.6
2020-11-27	0.0512	+0.3493	+149.2
2020-11-27	0.128	+0.3110	+75.5
2020-11-27	0.32	+0.1827	+14.2
2020-11-27	0.8	+0.2861	+237.0
2020-11-27	2	+0.2918	+51.5
2020-11-27	5	+0.4184	+241.6
2020-12-04	0.008192	+0.0599	+34.4

2020-12-04	0.02048	+0.3061	+25.6
2020-12-04	0.0512	+0.1850	+113.5
2020-12-04	0.128	+0.3075	+76.1
2020-12-04	0.32	+0.2394	+1.4
2020-12-04	0.8	+0.3160	+54.4
2020-12-04	2	+0.2506	+80.9
2020-12-04	5	+0.2039	+24.9

A1.3. Source patient 10-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2502; median raw-area contrast +60.43. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 24. A1 source patient 10-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-11-19	0.008192	+0.2087	+93.1
2020-11-19	0.02048	+0.1557	+69.1
2020-11-19	0.0512	+0.3545	+45.9
2020-11-19	0.128	+0.4348	+82.3
2020-11-19	0.32	+0.3166	+35.9
2020-11-19	0.8	+0.2305	+40.0
2020-11-19	2	+0.1722	+61.9
2020-11-19	5	+0.2287	+13.6
2020-11-25	0.008192	+0.1976	+20.9
2020-11-25	0.02048	+0.2663	+119.8
2020-11-25	0.0512	+0.3775	+122.3
2020-11-25	0.128	+0.4570	+123.5
2020-11-25	0.32	+0.3966	+50.4
2020-11-25	0.8	+0.4991	+84.5
2020-11-25	2	+0.3223	+1.1
2020-11-25	5	+0.1941	+25.3
2021-01-06	0.008192	+0.2091	+26.5
2021-01-06	0.02048	+0.2342	+95.2
2021-01-06	0.0512	+0.1848	+13.6
2021-01-06	0.128	+0.4305	+109.8
2021-01-06	0.32	+0.2299	+59.0
2021-01-06	0.8	+0.4818	+100.7
2021-01-06	2	+0.3073	-0.4
2021-01-06	5	+0.2228	+108.4

A1.4. Source patient 20-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.1534; median raw-area contrast +108.65. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 25. A1 source patient 20-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-27	0.008192	+0.1287	+135.3
2020-08-27	0.02048	+0.1368	+226.3
2020-08-27	0.0512	+0.1487	+129.1
2020-08-27	0.128	-0.0330	+123.2
2020-08-27	0.32	+0.1130	+171.5
2020-08-27	0.8	-0.1799	-12.5
2020-08-27	2	-0.0607	+49.3
2020-08-27	5	+0.0265	+133.3
2020-09-08	0.008192	+0.0891	-1.6
2020-09-08	0.02048	+0.1368	-13.8
2020-09-08	0.0512	+0.2866	+103.0
2020-09-08	0.128	+0.1152	+105.9
2020-09-08	0.32	+0.2096	+24.0
2020-09-08	0.8	+0.3125	+104.0
2020-09-08	2	+0.2647	+110.8
2020-09-08	5	+0.1582	-44.8
2020-09-15	0.008192	+0.1473	+97.1
2020-09-15	0.02048	+0.2262	+160.2
2020-09-15	0.0512	+0.3844	+157.1
2020-09-15	0.128	+0.3246	+191.7
2020-09-15	0.32	+0.3829	+133.5
2020-09-15	0.8	+0.3559	+106.5
2020-09-15	2	+0.3542	+14.2
2020-09-15	5	+0.1749	+155.8

A1.5. Source patient 21-CF

23 eligible of 24 blocks; median normalized treatment contrast +0.2527; median raw-area contrast +97.31. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 26. A1 source patient 21-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-13	0.008192	+0.1814	+62.9
2020-08-13	0.02048	+0.1679	-10.0
2020-08-13	0.0512	+0.3067	+97.3
2020-08-13	0.128	+0.4494	+200.0
2020-08-13	0.32	+0.9694	+126.1
2020-08-13	0.8	+0.5938	+69.2
2020-08-13	2	+0.7810	+220.4
2020-08-13	5	+0.8583	+217.4
2020-09-18	0.008192	+0.1843	-23.7
2020-09-18	0.02048	excluded	<2 wells/arm
2020-09-18	0.0512	+0.2527	+94.5
2020-09-18	0.128	+0.1219	+42.9
2020-09-18	0.32	+0.1303	-51.0
2020-09-18	0.8	+0.2095	+145.0
2020-09-18	2	+0.0519	+130.2
2020-09-18	5	+0.0882	-37.7
2020-09-24	0.008192	+0.2207	+127.9
2020-09-24	0.02048	+0.3608	+125.7
2020-09-24	0.0512	+0.2756	+113.1
2020-09-24	0.128	+0.2576	+229.0
2020-09-24	0.32	+0.3469	+42.5
2020-09-24	0.8	+0.3861	+109.3
2020-09-24	2	+0.1227	+38.6
2020-09-24	5	-0.0843	+59.5

A1.6. Source patient 22-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2165; median raw-area contrast +87.72. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 27. A1 source patient 22-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-27	0.008192	+0.1680	+61.0
2020-08-27	0.02048	+0.1146	+60.2
2020-08-27	0.0512	+0.2120	+191.8
2020-08-27	0.128	+0.3208	+128.2
2020-08-27	0.32	+0.3724	+219.8
2020-08-27	0.8	+0.3202	+94.8

2020-08-27	2	+0.3405	+190.0
2020-08-27	5	+0.0314	-137.4
2020-09-01	0.008192	+0.1006	+86.2
2020-09-01	0.02048	+0.1493	-4.5
2020-09-01	0.0512	+0.1727	+15.0
2020-09-01	0.128	+0.1303	+109.8
2020-09-01	0.32	+0.1500	+9.9
2020-09-01	0.8	+0.3940	+72.0
2020-09-01	2	+0.2823	+180.1
2020-09-01	5	+0.0797	+91.7
2020-09-24	0.008192	+0.1247	+87.8
2020-09-24	0.02048	+0.2495	+150.2
2020-09-24	0.0512	+0.2776	+49.7
2020-09-24	0.128	+0.2209	+87.7
2020-09-24	0.32	+0.2853	-67.6
2020-09-24	0.8	+0.4043	+241.1
2020-09-24	2	+0.1834	+51.3
2020-09-24	5	+0.2237	+104.2

A1.7. Source patient 41-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.3282; median raw-area contrast +71.25. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 28. A1 source patient 41-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-13	0.008192	+0.1753	+22.4
2020-08-13	0.02048	+0.2519	+114.0
2020-08-13	0.0512	+0.3822	+88.3
2020-08-13	0.128	+0.5059	+119.7
2020-08-13	0.32	+0.5738	+72.4
2020-08-13	0.8	+0.8446	+224.9
2020-08-13	2	+0.4962	+161.0
2020-08-13	5	+0.6016	+126.1
2020-08-20	0.008192	+0.1145	-1.7
2020-08-20	0.02048	+0.1807	+32.8
2020-08-20	0.0512	+0.2720	+32.3
2020-08-20	0.128	+0.4330	+60.2
2020-08-20	0.32	+0.5062	+93.3
2020-08-20	0.8	+0.5894	+128.8

2020-08-20	2	+0.3326	+50.8
2020-08-20	5	+0.3239	+7.4
2020-08-28	0.008192	+0.1596	+70.1
2020-08-28	0.02048	+0.1603	+49.8
2020-08-28	0.0512	+0.4934	+111.1
2020-08-28	0.128	+0.3831	+99.3
2020-08-28	0.32	+0.2534	+51.4
2020-08-28	0.8	+0.1989	-83.3
2020-08-28	2	+0.1937	-107.3
2020-08-28	5	+0.2143	+169.7

A1.8. Source patient 47-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.3002; median raw-area contrast +99.63. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 29. A1 source patient 47-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-27	0.008192	+0.2497	+99.1
2020-08-27	0.02048	+0.2530	+64.5
2020-08-27	0.0512	+0.3126	+74.7
2020-08-27	0.128	+0.3431	+135.7
2020-08-27	0.32	+0.5456	+165.3
2020-08-27	0.8	+0.2775	+42.5
2020-08-27	2	+0.4075	-12.4
2020-08-27	5	+0.3058	+217.4
2020-11-05	0.008192	+0.2353	+26.0
2020-11-05	0.02048	+0.1838	+133.4
2020-11-05	0.0512	+0.3257	+99.7
2020-11-05	0.128	+0.3508	+60.3
2020-11-05	0.32	+0.2867	+28.5
2020-11-05	0.8	+0.3139	+209.5
2020-11-05	2	+0.1556	+99.6
2020-11-05	5	+0.2898	+79.0
2020-11-24	0.008192	+0.0862	-34.1
2020-11-24	0.02048	+0.2555	+111.2
2020-11-24	0.0512	+0.4055	+182.2
2020-11-24	0.128	+0.2708	+97.8
2020-11-24	0.32	+0.3053	+105.9
2020-11-24	0.8	+0.3092	+123.7

2020-11-24	2	+0.3773	+187.2
2020-11-24	5	+0.2950	+148.1

A1.9. Source patient 57-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2752; median raw-area contrast +75.66. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 30. A1 source patient 57-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-27	0.008192	+0.1154	-51.2
2020-08-27	0.02048	+0.1659	+76.0
2020-08-27	0.0512	+0.2875	+75.3
2020-08-27	0.128	+0.3663	+186.3
2020-08-27	0.32	+0.3707	-19.4
2020-08-27	0.8	+0.0961	+33.6
2020-08-27	2	+0.5050	+375.7
2020-08-27	5	+0.5034	+273.6
2020-09-01	0.008192	+0.1516	+94.5
2020-09-01	0.02048	+0.2158	+137.0
2020-09-01	0.0512	+0.1955	+138.9
2020-09-01	0.128	+0.2320	+1.5
2020-09-01	0.32	+0.3394	+142.5
2020-09-01	0.8	+0.1961	+103.9
2020-09-01	2	+0.3137	+313.3
2020-09-01	5	+0.5433	+130.7
2020-09-08	0.008192	+0.1598	+8.1
2020-09-08	0.02048	+0.1472	+46.1
2020-09-08	0.0512	+0.3105	+72.0
2020-09-08	0.128	+0.2637	+65.4
2020-09-08	0.32	+0.2097	-37.6
2020-09-08	0.8	+0.2868	+61.1
2020-09-08	2	+0.3960	+120.1
2020-09-08	5	+0.3423	+36.4

A1.10. Source patient 58-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.0189; median raw-area contrast +33.19. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 31. A1 source patient 58-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-09-25	0.008192	-0.0068	-6.1
2020-09-25	0.02048	+0.1879	+163.8
2020-09-25	0.0512	+0.1724	+109.7
2020-09-25	0.128	+0.0598	+46.9
2020-09-25	0.32	-0.0104	-102.7
2020-09-25	0.8	-0.1529	+40.6
2020-09-25	2	-0.1628	-10.0
2020-09-25	5	-0.1372	+29.8
2020-10-13	0.008192	+0.1115	+89.5
2020-10-13	0.02048	+0.1405	+89.1
2020-10-13	0.0512	+0.0679	+7.2
2020-10-13	0.128	-0.0182	+49.3
2020-10-13	0.32	+0.3089	+95.3
2020-10-13	0.8	-0.1031	+19.0
2020-10-13	2	-0.0962	+14.2
2020-10-13	5	-0.0557	-1.8
2020-11-20	0.008192	+0.1027	-23.4
2020-11-20	0.02048	+0.0586	+42.0
2020-11-20	0.0512	+0.0210	+8.3
2020-11-20	0.128	+0.0302	+57.1
2020-11-20	0.32	+0.0169	-7.3
2020-11-20	0.8	-0.0293	+12.5
2020-11-20	2	+0.0856	+63.2
2020-11-20	5	-0.0563	+36.6

A1.11. Source patient 59-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.3739; median raw-area contrast +90.31. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 32. A1 source patient 59-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-13	0.008192	+0.1418	+45.9
2020-08-13	0.02048	+0.2441	+122.3
2020-08-13	0.0512	+0.2658	+89.4

2020-08-13	0.128	+0.5715	+81.5
2020-08-13	0.32	+0.5435	+19.1
2020-08-13	0.8	+0.6986	+121.4
2020-08-13	2	+0.5563	+75.5
2020-08-13	5	+0.4616	+87.6
2020-08-21	0.008192	+0.1856	+95.5
2020-08-21	0.02048	+0.2447	+67.3
2020-08-21	0.0512	+0.3450	+165.6
2020-08-21	0.128	+0.4972	+142.5
2020-08-21	0.32	+0.4028	+154.3
2020-08-21	0.8	+0.4609	+266.4
2020-08-21	2	+0.4136	+117.9
2020-08-21	5	+0.4991	+229.6
2020-08-28	0.008192	+0.1663	+44.8
2020-08-28	0.02048	+0.1817	+77.2
2020-08-28	0.0512	+0.1962	+91.2
2020-08-28	0.128	+0.3427	+56.7
2020-08-28	0.32	+0.1733	+86.7
2020-08-28	0.8	+0.4969	+146.9
2020-08-28	2	+0.5447	+204.8
2020-08-28	5	+0.2635	+4.4

A1.12. Source patient 60-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2855; median raw-area contrast +73.90. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 33. A1 source patient 60-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-28	0.008192	+0.0823	+25.4
2020-08-28	0.02048	+0.0338	+69.0
2020-08-28	0.0512	+0.2529	+12.3
2020-08-28	0.128	+0.2634	+79.1
2020-08-28	0.32	+0.2767	+98.3
2020-08-28	0.8	+0.3716	+62.2
2020-08-28	2	+0.2134	+90.7
2020-08-28	5	+0.3016	+58.9
2020-09-03	0.008192	+0.0974	+69.7
2020-09-03	0.02048	+0.2117	+118.4
2020-09-03	0.0512	+0.2263	+29.4

2020-09-03	0.128	+0.3349	+83.3
2020-09-03	0.32	+0.2943	+82.1
2020-09-03	0.8	+0.4011	+101.6
2020-09-03	2	+0.3729	+196.5
2020-09-03	5	+0.4666	+158.3
2021-02-17	0.008192	+0.1030	+50.1
2021-02-17	0.02048	+0.1163	+32.8
2021-02-17	0.0512	+0.2100	+55.7
2021-02-17	0.128	+0.3205	+59.3
2021-02-17	0.32	+0.4397	+165.4
2021-02-17	0.8	+0.3940	+107.2
2021-02-17	2	+0.3039	+66.6
2021-02-17	5	+0.3358	+78.1

A1.13. Source patient 61-CF

8 eligible of 8 blocks; median normalized treatment contrast +0.2065; median raw-area contrast +40.30. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 34. A1 source patient 61-CF: 8 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-10-14	0.008192	+0.1010	+3.4
2020-10-14	0.02048	+0.1303	+61.6
2020-10-14	0.0512	+0.2713	+94.0
2020-10-14	0.128	+0.3055	+1.0
2020-10-14	0.32	+0.1864	+62.9
2020-10-14	0.8	+0.2266	+9.0
2020-10-14	2	+0.2298	+33.6
2020-10-14	5	+0.1652	+47.0

A1.14. Source patient 62-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2456; median raw-area contrast +65.39. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 35. A1 source patient 62-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
------------	----------	------------------	------------------------

2020-10-13	0.008192	+0.1148	+59.2
2020-10-13	0.02048	+0.1510	+134.3
2020-10-13	0.0512	+0.2029	+99.3
2020-10-13	0.128	+0.2730	+3.1
2020-10-13	0.32	+0.2679	+112.1
2020-10-13	0.8	+0.3107	+190.4
2020-10-13	2	+0.2862	+198.4
2020-10-13	5	+0.2334	+35.7
2020-10-23	0.008192	+0.1637	+9.4
2020-10-23	0.02048	+0.2179	+164.0
2020-10-23	0.0512	+0.2163	+96.9
2020-10-23	0.128	+0.2786	+60.1
2020-10-23	0.32	+0.2270	+2.8
2020-10-23	0.8	+0.2539	+122.8
2020-10-23	2	+0.3275	+161.1
2020-10-23	5	+0.3026	-28.4
2020-11-03	0.008192	+0.1588	+70.7
2020-11-03	0.02048	+0.1511	+57.5
2020-11-03	0.0512	+0.1913	+56.3
2020-11-03	0.128	+0.2564	+169.0
2020-11-03	0.32	+0.2374	+118.7
2020-11-03	0.8	+0.2807	-52.1
2020-11-03	2	+0.2631	+14.4
2020-11-03	5	+0.2728	-7.6

A1.15. Source patient 63-CF

8 eligible of 8 blocks; median normalized treatment contrast +0.1083; median raw-area contrast +35.44. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 36. A1 source patient 63-CF: 8 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-10-14	0.008192	+0.0993	-54.9
2020-10-14	0.02048	+0.1585	+84.2
2020-10-14	0.0512	+0.1628	+56.2
2020-10-14	0.128	+0.0816	+34.7
2020-10-14	0.32	+0.0410	+29.7
2020-10-14	0.8	+0.1787	+36.2
2020-10-14	2	+0.0252	+4.6
2020-10-14	5	+0.1172	+49.2

A1.16. Source patient 64-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2636; median raw-area contrast +72.83. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 37. A1 source patient 64-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-10-02	0.008192	+0.1263	+68.9
2020-10-02	0.02048	+0.2773	+156.1
2020-10-02	0.0512	+0.3902	+61.7
2020-10-02	0.128	+0.3963	-61.1
2020-10-02	0.32	+0.3450	+190.4
2020-10-02	0.8	+0.1578	+87.4
2020-10-02	2	+0.8928	+411.1
2020-10-02	5	+0.1912	+28.6
2020-10-13	0.008192	+0.1390	+10.9
2020-10-13	0.02048	+0.1920	+52.1
2020-10-13	0.0512	+0.2484	+132.3
2020-10-13	0.128	+0.4522	+160.1
2020-10-13	0.32	+0.3708	+94.7
2020-10-13	0.8	+0.3116	+132.1
2020-10-13	2	+0.2478	+76.8
2020-10-13	5	+0.1562	+62.1
2020-10-23	0.008192	+0.1398	+36.0
2020-10-23	0.02048	+0.1458	+45.2
2020-10-23	0.0512	+0.2500	+7.5
2020-10-23	0.128	+0.3986	+145.2
2020-10-23	0.32	+0.2824	+18.9
2020-10-23	0.8	+0.4767	+133.5
2020-10-23	2	+0.6337	+302.3
2020-10-23	5	+0.1829	-32.5

A1.17. Source patient 65-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2475; median raw-area contrast +57.82. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 38. A1 source patient 65-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-10-15	0.008192	+0.1919	+15.4
2020-10-15	0.02048	+0.1633	+42.4
2020-10-15	0.0512	+0.2534	+101.4
2020-10-15	0.128	+0.1381	-37.9
2020-10-15	0.32	+0.1599	+61.3
2020-10-15	0.8	+0.1846	+4.7
2020-10-15	2	+0.2454	+94.4
2020-10-15	5	+0.2965	+69.6
2020-10-23	0.008192	+0.1632	-3.8
2020-10-23	0.02048	+0.2079	+87.6
2020-10-23	0.0512	+0.2214	+19.4
2020-10-23	0.128	+0.3708	+148.7
2020-10-23	0.32	+0.4989	+141.1
2020-10-23	0.8	+0.2196	-36.8
2020-10-23	2	+0.2531	+73.6
2020-10-23	5	+0.2677	+95.7
2020-11-05	0.008192	+0.1974	+23.7
2020-11-05	0.02048	+0.2906	+92.2
2020-11-05	0.0512	+0.1865	+38.7
2020-11-05	0.128	+0.3228	+154.6
2020-11-05	0.32	+0.2499	+54.3
2020-11-05	0.8	+0.4921	+62.0
2020-11-05	2	+0.2495	+10.8
2020-11-05	5	+0.2790	+9.3

A1.18. Source patient 66-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2951; median raw-area contrast +84.77. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 39. A1 source patient 66-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-10-15	0.008192	+0.1924	+12.4
2020-10-15	0.02048	+0.2020	+46.6
2020-10-15	0.0512	+0.2868	+56.2
2020-10-15	0.128	+0.3330	+36.3
2020-10-15	0.32	+0.2706	+94.5
2020-10-15	0.8	+0.1987	+63.7

2020-10-15	2	+0.1706	+45.6
2020-10-15	5	+0.3330	+68.4
2020-10-27	0.008192	+0.1634	+66.3
2020-10-27	0.02048	+0.2660	+90.9
2020-10-27	0.0512	+0.3031	+113.0
2020-10-27	0.128	+0.3836	+39.4
2020-10-27	0.32	+0.4641	+137.8
2020-10-27	0.8	+0.2784	+53.5
2020-10-27	2	+0.2882	+82.8
2020-10-27	5	+0.3812	+139.7
2020-11-05	0.008192	+0.1517	+105.6
2020-11-05	0.02048	+0.3261	+151.9
2020-11-05	0.0512	+0.3985	+114.8
2020-11-05	0.128	+0.3465	+14.0
2020-11-05	0.32	+0.4124	+154.4
2020-11-05	0.8	+0.3754	+192.8
2020-11-05	2	+0.3020	+100.9
2020-11-05	5	+0.1208	+86.7

A1.19. Source patient 68-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2833; median raw-area contrast +96.63. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 40. A1 source patient 68-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-10-21	0.008192	+0.1733	+76.5
2020-10-21	0.02048	+0.2356	+90.2
2020-10-21	0.0512	+0.3633	+172.9
2020-10-21	0.128	+0.4029	+17.3
2020-10-21	0.32	+0.3608	+120.1
2020-10-21	0.8	+0.3152	+205.5
2020-10-21	2	+0.2931	+103.5
2020-10-21	5	+0.3547	+103.1
2020-10-27	0.008192	+0.1400	+69.6
2020-10-27	0.02048	+0.1795	+125.8
2020-10-27	0.0512	+0.2906	+124.6
2020-10-27	0.128	+0.3525	+148.7
2020-10-27	0.32	+0.1431	+6.5
2020-10-27	0.8	+0.3659	+215.5

2020-10-27	2	+0.2594	+176.9
2020-10-27	5	-0.0718	-87.1
2020-11-27	0.008192	+0.1909	-4.2
2020-11-27	0.02048	+0.2265	+82.5
2020-11-27	0.0512	+0.3456	+52.2
2020-11-27	0.128	+0.4028	+153.4
2020-11-27	0.32	+0.2824	+58.8
2020-11-27	0.8	+0.2157	+68.5
2020-11-27	2	+0.1461	+27.8
2020-11-27	5	+0.2842	+156.9

A1.20. Source patient 69-CF

8 eligible of 8 blocks; median normalized treatment contrast +0.1718; median raw-area contrast +52.84. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 41. A1 source patient 69-CF: 8 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-10-23	0.008192	+0.1478	-6.2
2020-10-23	0.02048	+0.1696	+17.8
2020-10-23	0.0512	+0.1919	+47.8
2020-10-23	0.128	+0.2667	+81.7
2020-10-23	0.32	+0.1560	-23.6
2020-10-23	0.8	+0.1740	+139.8
2020-10-23	2	+0.2589	+57.9
2020-10-23	5	+0.0735	+80.4

A2. VTI minus ETI

Source <https://zenodo.org/records/15754800>. 12 IDs and 33 plates appear in the public source; 208/264 blocks meet its preregistered primary criteria. The table preserves positive and negative contrasts. The registered patient-level result is 6/9 positive medians, exact one-sided $p=0.2539062$. Unnormalized raw-area AUC is shown only to reveal sensitivity to baseline area, never as an independent trial.

A2.1. Source patient 06

24 eligible of 24 blocks; median normalized treatment contrast +0.0277; median raw-area contrast -8.49. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 42. A2 source patient 06: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240409	0.008192	+0.0673	+14.6
20240409	0.02048	+0.1224	+125.7
20240409	0.0512	+0.1283	-8.1
20240409	0.128	+0.0963	+56.1
20240409	0.32	-0.2065	+39.9
20240409	0.8	-0.0746	-33.6
20240409	2	+0.0106	+119.2
20240409	5	-0.1393	-49.6
20240416	0.008192	+0.1074	-24.7
20240416	0.02048	+0.0635	+80.1
20240416	0.0512	-0.0403	+16.7
20240416	0.128	+0.0839	-19.4
20240416	0.32	-0.0454	-26.3
20240416	0.8	+0.0135	-8.8
20240416	2	-0.1746	-47.1
20240416	5	-0.0725	-24.5
20240430	0.008192	+0.0397	-30.6
20240430	0.02048	+0.0156	+38.1
20240430	0.0512	+0.0455	-112.0
20240430	0.128	+0.0783	-134.1
20240430	0.32	-0.0534	+50.5
20240430	0.8	-0.0909	+29.6
20240430	2	+0.1366	-110.9
20240430	5	+0.2357	+135.0

A2.2. Source patient 27

24 eligible of 24 blocks; median normalized treatment contrast +0.0449; median raw-area contrast -4.67. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 43. A2 source patient 27: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240409	0.008192	+0.1404	+19.1
20240409	0.02048	+0.1543	+20.4
20240409	0.0512	+0.1201	+8.6

20240409	0.128	+0.2176	+48.1
20240409	0.32	-0.0955	-56.9
20240409	0.8	-0.0967	-102.1
20240409	2	-0.1990	+33.4
20240409	5	-0.0490	-5.1
20240416	0.008192	+0.0569	-1.1
20240416	0.02048	+0.0256	-13.5
20240416	0.0512	-0.0417	-40.9
20240416	0.128	+0.1041	-55.2
20240416	0.32	+0.0610	-17.7
20240416	0.8	-0.1142	+19.5
20240416	2	+0.1909	-4.2
20240416	5	-0.0018	-53.3
20240430	0.008192	+0.0236	+123.2
20240430	0.02048	-0.0882	+35.6
20240430	0.0512	+0.0329	-43.1
20240430	0.128	+0.1594	-49.5
20240430	0.32	+0.1445	-29.3
20240430	0.8	+0.2277	+26.0
20240430	2	-0.0951	-138.9
20240430	5	+0.0678	+96.4

A2.3. Source patient 31

24 eligible of 24 blocks; median normalized treatment contrast -0.0071; median raw-area contrast +1.27. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 44. A2 source patient 31: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240517	0.008192	+0.0205	+56.5
20240517	0.02048	-0.0042	+46.6
20240517	0.0512	-0.0206	-9.6
20240517	0.128	-0.0734	+20.9
20240517	0.32	-0.0520	-86.1
20240517	0.8	-0.0064	+26.0
20240517	2	-0.0664	-11.4
20240517	5	+0.0476	+23.2
20240523	0.008192	-0.0078	+6.4
20240523	0.02048	+0.0407	-95.6
20240523	0.0512	+0.0226	+136.1

20240523	0.128	-0.0575	-30.9
20240523	0.32	+0.0873	-109.2
20240523	0.8	-0.0168	+74.6
20240523	2	+0.2221	+129.3
20240523	5	-0.0983	-144.8
20240530	0.008192	-0.0413	-0.1
20240530	0.02048	+0.0166	-1.0
20240530	0.0512	-0.0805	-21.8
20240530	0.128	+0.0045	+2.6
20240530	0.32	+0.0143	-10.7
20240530	0.8	-0.0503	+54.3
20240530	2	+0.0192	+28.2
20240530	5	-0.2074	-43.3

A2.4. Source patient 35

No eligible primary block for this ID under the fixed treatment arms and time/well criteria. 16 source blocks are retained below as exclusions; this ID contributes no patient median or sign-test observation.

Table 45. A2 source patient 35: 16 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240314	0.008192	excluded	missing matched times
20240314	0.02048	excluded	missing matched times
20240314	0.0512	excluded	missing matched times
20240314	0.128	excluded	missing matched times
20240314	0.32	excluded	missing matched times
20240314	0.8	excluded	missing matched times
20240314	2	excluded	missing matched times
20240314	5	excluded	missing matched times
20240326	0.008192	excluded	missing matched times
20240326	0.02048	excluded	missing matched times
20240326	0.0512	excluded	missing matched

			times
20240326	0.128	excluded	missing matched times
20240326	0.32	excluded	missing matched times
20240326	0.8	excluded	missing matched times
20240326	2	excluded	missing matched times
20240326	5	excluded	missing matched times

A2.5. Source patient 40

16 eligible of 24 blocks; median normalized treatment contrast -0.0074; median raw-area contrast +3.17. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 46. A2 source patient 40: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20230627	0.008192	excluded	missing matched times
20230627	0.02048	excluded	missing matched times
20230627	0.0512	excluded	missing matched times
20230627	0.128	excluded	missing matched times
20230627	0.32	excluded	missing matched times
20230627	0.8	excluded	missing matched times
20230627	2	excluded	missing matched times
20230627	5	excluded	missing matched times
20241121	0.008192	+0.0078	-21.6
20241121	0.02048	+0.0141	-38.6
20241121	0.0512	-0.1030	+22.9
20241121	0.128	-0.0426	-146.0
20241121	0.32	+0.2843	-188.0
20241121	0.8	-0.3458	-179.2

20241121	2	-0.0569	-260.8
20241121	5	+0.1733	+14.7
20241204	0.008192	+0.0137	+13.7
20241204	0.02048	+0.0022	+20.5
20241204	0.0512	-0.0329	+74.1
20241204	0.128	-0.0665	-165.2
20241204	0.32	-0.0024	+71.5
20241204	0.8	-0.0125	+286.5
20241204	2	-0.0851	+134.1
20241204	5	+0.0549	-7.3

A2.6. Source patient 45

No eligible primary block for this ID under the fixed treatment arms and time/well criteria. 16 source blocks are retained below as exclusions; this ID contributes no patient median or sign-test observation.

Table 47. A2 source patient 45: 16 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240314	0.008192	excluded	missing matched times
20240314	0.02048	excluded	missing matched times
20240314	0.0512	excluded	missing matched times
20240314	0.128	excluded	missing matched times
20240314	0.32	excluded	missing matched times
20240314	0.8	excluded	missing matched times
20240314	2	excluded	missing matched times
20240314	5	excluded	missing matched times
20240328	0.008192	excluded	missing matched times
20240328	0.02048	excluded	missing matched times
20240328	0.0512	excluded	missing matched times
20240328	0.128	excluded	missing matched times

20240328	0.32	excluded	missing matched times
20240328	0.8	excluded	missing matched times
20240328	2	excluded	missing matched times
20240328	5	excluded	missing matched times

A2.7. Source patient 46

24 eligible of 24 blocks; median normalized treatment contrast +0.0345; median raw-area contrast -2.46. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 48. A2 source patient 46: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240523	0.008192	-0.0261	-1.5
20240523	0.02048	-0.0601	+35.9
20240523	0.0512	+0.0294	-3.4
20240523	0.128	+0.1802	+39.0
20240523	0.32	-0.1560	-95.0
20240523	0.8	+0.0617	+59.8
20240523	2	-0.1395	+5.8
20240523	5	+0.1327	+28.1
20240530	0.008192	+0.0655	-28.3
20240530	0.02048	-0.0300	-21.6
20240530	0.0512	+0.1325	+76.6
20240530	0.128	+0.0425	-12.1
20240530	0.32	+0.1158	-20.2
20240530	0.8	-0.1137	+82.9
20240530	2	+0.0750	+129.5
20240530	5	-0.0437	-38.0
20240625	0.008192	+0.0610	+126.1
20240625	0.02048	+0.0397	-132.9
20240625	0.0512	+0.1187	-141.3
20240625	0.128	-0.1928	-28.4
20240625	0.32	-0.1743	-94.4
20240625	0.8	-0.1692	-365.7
20240625	2	+0.0176	+43.8
20240625	5	+0.3418	+282.1

A2.8. Source patient 58

No eligible primary block for this ID under the fixed treatment arms and time/well criteria. 16 source blocks are retained below as exclusions; this ID contributes no patient median or sign-test observation.

Table 49. A2 source patient 58: 16 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240314	0.008192	excluded	missing matched times
20240314	0.02048	excluded	missing matched times
20240314	0.0512	excluded	missing matched times
20240314	0.128	excluded	missing matched times
20240314	0.32	excluded	missing matched times
20240314	0.8	excluded	missing matched times
20240314	2	excluded	missing matched times
20240314	5	excluded	missing matched times
20241125	0.008192	excluded	missing matched times
20241125	0.02048	excluded	missing matched times
20241125	0.0512	excluded	missing matched times
20241125	0.128	excluded	missing matched times
20241125	0.32	excluded	missing matched times
20241125	0.8	excluded	missing matched times
20241125	2	excluded	missing matched times
20241125	5	excluded	missing matched times

A2.9. Source patient 68

24 eligible of 24 blocks; median normalized treatment contrast -0.0086; median raw-area contrast -6.57. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 50. A2 source patient 68: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240523	0.008192	+0.1833	+10.7
20240523	0.02048	-0.0085	-19.6
20240523	0.0512	-0.0683	+104.4
20240523	0.128	-0.0599	+6.4
20240523	0.32	-0.0783	+32.1
20240523	0.8	-0.1295	-111.1
20240523	2	-0.0366	+31.4
20240523	5	-0.0529	-222.3
20240618	0.008192	+0.0920	-164.7
20240618	0.02048	+0.0920	+34.0
20240618	0.0512	-0.0680	-83.5
20240618	0.128	+0.2306	+205.8
20240618	0.32	+0.0186	+15.5
20240618	0.8	+0.0648	+58.2
20240618	2	-0.0087	+118.5
20240618	5	+0.0629	+198.5
20240625	0.008192	+0.0275	-111.6
20240625	0.02048	+0.1288	-50.9
20240625	0.0512	+0.0323	-112.7
20240625	0.128	+0.0999	+86.8
20240625	0.32	-0.1190	-175.5
20240625	0.8	-0.0566	-151.4
20240625	2	-0.3497	-407.8
20240625	5	-0.1398	-215.7

A2.10. Source patient 91

24 eligible of 24 blocks; median normalized treatment contrast +0.0282; median raw-area contrast -18.70. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 51. A2 source patient 91: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240411	0.008192	+0.0229	-6.6
20240411	0.02048	-0.0303	-14.2
20240411	0.0512	-0.1336	-142.0
20240411	0.128	+0.0830	-23.6
20240411	0.32	-0.0511	-60.4
20240411	0.8	-0.0090	-48.8
20240411	2	+0.1982	+28.7
20240411	5	-0.0349	-25.5
20240418	0.008192	+0.0667	+4.1
20240418	0.02048	+0.0713	+34.1
20240418	0.0512	-0.0999	+44.8
20240418	0.128	+0.1409	-7.5
20240418	0.32	-0.0271	-12.9
20240418	0.8	-0.3972	-113.6
20240418	2	-0.1867	-130.8
20240418	5	+0.2601	+74.7
20240530	0.008192	+0.0539	-58.3
20240530	0.02048	+0.0612	-89.6
20240530	0.0512	+0.2280	+88.2
20240530	0.128	-0.0384	-79.0
20240530	0.32	+0.0473	-23.2
20240530	0.8	+0.0482	+14.0
20240530	2	+0.0335	+87.7
20240530	5	-0.0604	-64.9

A2.11. Source patient 94

24 eligible of 24 blocks; median normalized treatment contrast +0.0181; median raw-area contrast -1.68. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 52. A2 source patient 94: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240517	0.008192	-0.0207	+32.3
20240517	0.02048	+0.1247	-15.3
20240517	0.0512	+0.1258	+37.5
20240517	0.128	+0.0476	+63.3
20240517	0.32	-0.0145	-81.3
20240517	0.8	+0.0029	-5.6

20240517	2	+0.0201	-212.3
20240517	5	-0.2070	-9.6
20240523	0.008192	+0.0776	-5.9
20240523	0.02048	-0.0184	+2.2
20240523	0.0512	-0.0364	+111.8
20240523	0.128	+0.0162	-11.8
20240523	0.32	-0.1115	+2.4
20240523	0.8	-0.0523	+17.8
20240523	2	-0.0246	+79.5
20240523	5	+0.0761	-146.5
20240618	0.008192	+0.0082	+151.4
20240618	0.02048	+0.1450	-247.8
20240618	0.0512	+0.0795	+27.1
20240618	0.128	+0.1825	+126.8
20240618	0.32	+0.0732	-34.9
20240618	0.8	-0.0403	-20.6
20240618	2	+0.0750	+165.2
20240618	5	+0.0522	-131.9

A2.12. Source patient 96

24 eligible of 24 blocks; median normalized treatment contrast +0.0039; median raw-area contrast +14.80. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 53. A2 source patient 96: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240510	0.008192	-0.0213	+22.4
20240510	0.02048	-0.0038	+36.6
20240510	0.0512	+0.0183	-58.3
20240510	0.128	+0.1031	+11.8
20240510	0.32	-0.0187	-26.2
20240510	0.8	+0.1301	+35.0
20240510	2	+0.0947	+88.4
20240510	5	-0.0399	+22.1
20240517	0.008192	+0.0189	+20.4
20240517	0.02048	-0.0141	-19.0
20240517	0.0512	+0.0410	+2.9
20240517	0.128	-0.0070	-30.7
20240517	0.32	-0.0814	+95.5
20240517	0.8	-0.1982	+129.7

20240517	2	+0.0774	+17.8
20240517	5	+0.0117	+9.6
20240620	0.008192	+0.1536	+5.1
20240620	0.02048	+0.0341	-50.5
20240620	0.0512	+0.1347	+79.0
20240620	0.128	+0.0341	-47.5
20240620	0.32	-0.1029	+18.5
20240620	0.8	-0.2127	-150.3
20240620	2	-0.0377	-122.1
20240620	5	-0.1185	+44.4

Audit boundary. A plate-dose block is not an independent donor. Within each patient the median across eligible blocks gives one observation for the exact sign test, preventing a large plate or many concentrations from dominating the patient count. Raw-area reversals flag sensitivity to baseline organoid density. For excluded A2 patient blocks, a missing default treatment arm is not replaced with a different VTI concentration. This appendix is a reproducibility trace of two already published in-vitro comparisons, not evidence for clinical benefit, equivalence, or the single-object size candidate.